

## The topic is

- A topic is a subject or a theme that is discussed or written about.
- A topic can be general or specific, depending on the purpose and scope of the discussion or writing.
- A topic can be chosen by the writer or speaker, or assigned by a teacher or an editor.
- A topic can be expressed as a word, a phrase, a question, or a statement.
- A topic can be related to various fields of study, such as science, history, literature, art, etc.
- A topic can be informative, persuasive, argumentative, descriptive, narrative, or analytical, depending on the type and goal of the writing or speaking.
- A topic can be narrowed down or broadened by using subtopics, examples, details, evidence, or arguments.
- A topic can be organized by using an outline, a mind map, a graphic organizer, or a thesis statement.
- A topic can be developed by using research, facts, opinions, examples, anecdotes, statistics, quotations, or personal experiences.
- A topic can be revised by checking for clarity, coherence, relevance, accuracy, completeness, and originality.

## FUNDAMENTALS OF ELECTRONICS ENGINEERING

- Electronics engineering is a branch of engineering that deals with the design, development and testing of electronic systems and products that use digital or analog components.
- Electronic systems and products can be used for various applications such as telecommunications, science, healthcare, government, and military.
- Electronics engineering requires a solid understanding of the basic concepts and principles of electricity, circuits, components, and devices.
- Some of the basic concepts and principles of electronics engineering are:
  - Electricity: Electricity is the flow of electric charge through a conductor. There are two types of electrical signals, those being alternating current (AC), and direct current (DC). AC is the type of electricity that comes from the wall outlet and changes direction periodically. DC is the type of electricity that comes from batteries and flows in one direction only.
  - Circuits: A circuit is a complete and closed path through which electric current can flow. A circuit consists of a power source, such as a battery or a generator, and one or more components, such as

resistors, capacitors, diodes, transistors, etc., that are connected by wires or traces.

- Resistance: Resistance is the measure of how much a component or a material opposes the flow of electric current. Resistance is measured in ohms ( $\Omega$ ) and is calculated by dividing the voltage (V) across the component by the current (I) through it. Resistance can be used to control the amount of current in a circuit or to create voltage drops.
- Series Vs. Parallel: Series and parallel are two ways of connecting components in a circuit. In a series connection, the components are connected end to end, so that the same current flows through all of them. The total resistance of a series connection is the sum of the individual resistances. In a parallel connection, the components are connected across each other, so that the same voltage is applied to all of them. The total resistance of a parallel connection is the reciprocal of the sum of the reciprocals of the individual resistances.
- Basic Components: Some of the basic components used in electronics engineering are:
  - \* Resistors: Resistors are components that have a fixed or variable resistance. Resistors are used to limit the current, divide the voltage, or create heat in a circuit. Resistors are color-coded to indicate their resistance value and tolerance.
  - \* Capacitors: Capacitors are components that store electric charge and energy. Capacitors are used to filter, smooth, or store voltage, or to create timing or frequency effects in a circuit. Capacitors are labeled with their capacitance value and voltage rating.
  - \* Diodes: Diodes are components that allow current to flow in one direction only. Diodes are used to rectify, regulate, or protect voltage, or to create light or sound in a circuit. Diodes are marked with a stripe or a band to indicate their polarity.
  - \* Transistors: Transistors are components that can amplify or switch electric signals. Transistors are used to create logic, memory, or power functions in a circuit. Transistors have three terminals: the base, the collector, and the emitter, and are identified by their type and number.

- These are some of the fundamentals of electronics engineering. For more details, please refer to the sources below.

: <https://www.instructables.com/Basic-Electronics/> <https://www.indeed.com/career-advice/finding-a-job/what-is-electronics-engineering>  
<https://www.edx.org/learn/electronics>

## Unit 1 - Semiconductor Diode

- A semiconductor diode is a device that allows current to flow in one direction, but blocks it in the opposite direction.
- A semiconductor diode is made of two types of semiconductor materials: p-type and n-type.
- The p-type semiconductor has a lot of holes, which are positively charged carriers, while the n-type semiconductor has a lot of electrons, which are negatively charged carriers.
- The junction between the p-type and n-type semiconductors is called the pn junction.
- The pn junction has two terminals: the anode, which is connected to the p-type side, and the cathode, which is connected to the n-type side.
- The symbol for a semiconductor diode is shown below:

diode symbol

- The arrow in the symbol indicates the direction of the forward current, which is the current that flows when the anode is more positive than the cathode.
- The vertical line in the symbol represents the reverse bias, which is the condition when the anode is more negative than the cathode.
- In reverse bias, the diode blocks the current flow, except for a very small leakage current.
- The diode has a characteristic called the forward voltage drop, which is the minimum voltage difference between the anode and the cathode that is required for the diode to conduct current in the forward direction.
- The forward voltage drop depends on the type of semiconductor material used in the diode. For example, silicon diodes have a forward voltage drop of about 0.7 V, while germanium diodes have a forward voltage drop of about 0.3 V.
- The diode also has a characteristic called the reverse breakdown voltage, which is the maximum voltage difference between the anode and the cathode that the diode can withstand in the reverse direction without being damaged.
- The reverse breakdown voltage depends on the design and construction of the diode. For example, Zener diodes are specially designed to have a low reverse breakdown voltage, which makes them useful for voltage regulation.
- The diode can be used for various applications, such as rectification, switching, clamping, clipping, modulation, and protection.
- Rectification is the process of converting alternating current (AC) into direct current (DC) by using one or more diodes.
- Switching is the process of controlling the flow of current in a circuit by using a diode as a switch.
- Clamping is the process of shifting the voltage level of a signal by using a diode and a capacitor.

- Clipping is the process of limiting the amplitude of a signal by using a diode and a resistor.
- Modulation is the process of changing the characteristics of a carrier signal by using a diode and a modulating signal.
- Protection is the process of preventing damage to a circuit or a device by using a diode as a safeguard.

### **Depletion layer**

- A depletion layer is an insulating region within a conductive, doped semiconductor material where the mobile charge carriers have been diffused away, or have been forced away by an electric field.
- A depletion layer forms when a p-type semiconductor is joined with an n-type semiconductor, creating a p-n junction.
- The p-type semiconductor has a higher concentration of holes, while the n-type semiconductor has a higher concentration of electrons.
- Due to the difference in concentration, holes diffuse from the p-region to the n-region, and electrons diffuse from the n-region to the p-region, causing recombination.
- As a result, a thin layer near the junction becomes depleted of charge carriers, leaving behind fixed and oppositely charged ions.
- The depletion layer acts as a barrier that opposes the flow of electrons from the n-side to the p-side, and vice versa.
- The depletion layer also creates an electric field that points from the n-side to the p-side, which further prevents the diffusion of charge carriers.
- The width and potential of the depletion layer depend on the doping levels of the semiconductors, the applied voltage, and the temperature.
- The depletion layer plays an important role in the operation of semiconductor devices, such as diodes, transistors, solar cells, and light-emitting diodes.

### **V-I characteristics of semiconductor diode**

- The V-I characteristics of a semiconductor diode is a curve that shows the relationship between the voltage across the diode and the current through it.
- The V-I characteristics can be obtained by connecting a diode in series with a resistor and a variable voltage source, as shown in the figure below.

Circuit diagram for V-I characteristics of diode

- The V-I characteristics can be divided into two regions: forward bias and reverse bias.
- Forward bias is the condition when the positive terminal of the voltage source is connected to the p-type side of the diode and the negative terminal to the n-type side. In this case, the diode offers a low resistance and allows a large current to flow through it.

- Reverse bias is the condition when the positive terminal of the voltage source is connected to the n-type side of the diode and the negative terminal to the p-type side. In this case, the diode offers a high resistance and allows a very small current to flow through it.
- The V-I characteristics curve for forward and reverse bias is shown in the figure below.

#### V-I characteristics curve for diode

- The forward bias region of the curve has three parts: the cut-in voltage, the linear region, and the saturation region.
- The cut-in voltage is the minimum voltage required to make the diode conduct in the forward direction. It is also called the threshold voltage or the knee voltage. It depends on the type and doping level of the semiconductor material. For silicon diodes, it is about 0.7 V and for germanium diodes, it is about 0.3 V.
- The linear region is the part of the curve where the current increases linearly with the voltage. The slope of the curve in this region is the forward resistance of the diode, which is usually very small.
- The saturation region is the part of the curve where the current reaches a maximum value and does not increase further with the voltage. This is because the diode cannot handle more current without overheating or breaking down. The voltage at this point is called the forward voltage or the forward drop of the diode.
- The reverse bias region of the curve has two parts: the leakage current and the breakdown region.
- The leakage current is the very small current that flows through the diode in the reverse direction. It is due to the thermal generation of electron-hole pairs in the depletion region of the diode. It is usually negligible and can be ignored in most applications.
- The breakdown region is the part of the curve where the current increases rapidly with the voltage in the reverse direction. This is because the diode undergoes a phenomenon called breakdown, which is the breakdown of the depletion region due to a high electric field. There are two types of breakdown: avalanche breakdown and Zener breakdown. Avalanche breakdown occurs when the electrons gain enough kinetic energy to knock out more electrons from the atoms, creating a chain reaction of ionization. Zener breakdown occurs when the electric field is so strong that it pulls out electrons from the valence band, creating a large number of electron-hole pairs. Both types of breakdown result in a large current flow through the diode, which can damage it if not controlled. The voltage at which breakdown occurs is called the breakdown voltage or the Zener voltage of the diode. It depends on the doping level and the thickness of the depletion region of the diode. Some diodes are specially designed to operate in the breakdown region and are called Zener diodes. They are used for voltage regulation and reference purposes.

## Ideal and Practical Diodes

- A diode is a two-terminal electronic device that allows current to flow in one direction only.
- An ideal diode is a hypothetical device that has zero resistance when forward biased and infinite resistance when reverse biased.
- A practical diode is a real device that has some non-ideal characteristics, such as voltage drop, leakage current, and breakdown voltage.
- The main differences between ideal and practical diodes are:

| Ideal diodes                                                 | Practical diodes                                                                        |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Ideal diodes act as perfect conductor and perfect insulator. | Practical diodes cannot act as perfect conductor or insulator.                          |
| Ideal diode draws no current when reverse biased.            | Practical diode draws very low current when reverse biased, called leakage current.     |
| Ideal diode offers infinite resistance when reverse biased.  | Practical diode offers very high resistance when reverse biased, but not infinite.      |
| Ideal diode has no voltage drop when forward biased.         | Practical diode has a voltage drop when forward biased, called forward voltage.         |
| Ideal diode has no breakdown voltage.                        | Practical diode has a breakdown voltage, beyond which it conducts in reverse direction. |
| Ideal diode cannot be manufactured.                          | Practical diode can be manufactured.                                                    |

- The V-I characteristics of ideal and practical diodes are shown in the figure below:

V-I characteristics of ideal and practical diodes

- An ideal diode can be used as a switch, as it can turn on and off the current flow depending on the polarity of the applied voltage.
- A practical diode can be used for various applications, such as rectification, clipping, clamping, modulation, switching, and logic gates.

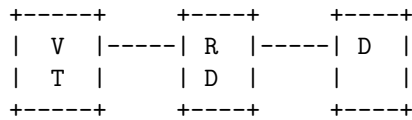
## Diode Equivalent Circuits

- An equivalent circuit is a combination of elements that best represents the actual terminal characteristics of the device.
- An equivalent circuit can be used to simplify the analysis of a circuit containing a diode, by replacing the diode with other elements without severely affecting the behavior of the circuit.
- There are different types of equivalent circuits for a diode, depending on the level of accuracy and complexity required.

- Three models with increasing accuracy are listed below:

#### 1. Piecewise-Linear Equivalent Circuit

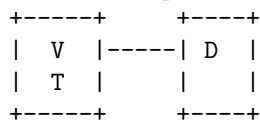
- A technique for obtaining an equivalent circuit for a diode is to approximate the characteristics of the device by straight-line segments.
- The resulting equivalent circuit is naturally called the piecewise-linear equivalent circuit.
- The piecewise-linear equivalent circuit consists of a voltage source, a resistor, and an ideal diode.
- The voltage source represents the threshold voltage of the diode, which is the minimum voltage required to turn on the diode.
- The resistor represents the dynamic resistance of the diode, which is the slope of the diode characteristic curve near the operating point.
- The ideal diode is a switch that is either open or closed, depending on the polarity of the applied voltage.
- The piecewise-linear equivalent circuit is shown below:



- Where  $V_T$  is the threshold voltage,  $R_D$  is the dynamic resistance, and  $D$  is the ideal diode.
- The piecewise-linear equivalent circuit is useful for analyzing circuits with small variations in voltage and current around a given operating point.
- However, it is not accurate for large variations or for different types of diodes.

#### 2. Simplified Equivalent Circuit

- The equivalent model in this case consists of a battery and an ideal diode.
- The battery represents the threshold voltage of the diode, and the ideal diode is the same as before.
- The simplified equivalent circuit is shown below:



- Where  $V_T$  is the threshold voltage, and  $D$  is the ideal diode.
- The simplified equivalent circuit is useful for analyzing circuits with large variations in voltage and current, or for different types of diodes.
- However, it ignores the dynamic resistance of the diode, which may affect the circuit performance.

#### 3. Ideal Diode Model

- The equivalent model in this case consists of only an ideal diode.
- The ideal diode is a switch that is either open or closed, depending on the polarity of the applied voltage.

- The ideal diode model is shown below:



- Where D is the ideal diode.
- The ideal diode model is useful for analyzing circuits with very large variations in voltage and current, or for qualitative understanding of the diode behavior.
- However, it ignores the threshold voltage and the dynamic resistance of the diode, which may affect the circuit performance.

### **Zener Diodes breakdown mechanism (Zener and avalanche)**

- A Zener diode is a special type of diode that can operate in the reverse breakdown region without being damaged.
- The reverse breakdown region is the region where the reverse voltage across the diode exceeds a certain value, called the breakdown voltage, and a large reverse current flows through the diode.
- There are two types of breakdown mechanisms that can occur in a Zener diode: Zener breakdown and avalanche breakdown.
- Zener breakdown occurs when the reverse voltage is low (typically less than 5 V) and the electric field across the depletion region is high enough to break the covalent bonds of the semiconductor atoms and generate electron-hole pairs. These carriers contribute to the reverse current and lower the resistance of the diode.
- Avalanche breakdown occurs when the reverse voltage is high (typically more than 5 V) and the electric field across the depletion region accelerates the minority carriers to a high velocity, causing them to collide with the lattice atoms and create more electron-hole pairs. These carriers also contribute to the reverse current and lower the resistance of the diode.
- The breakdown voltage of a Zener diode depends on the doping concentration and the thickness of the depletion region. A higher doping concentration and a thinner depletion region result in a lower breakdown voltage and a higher reverse current.
- The breakdown voltage of a Zener diode is also temperature dependent. Zener breakdown voltage decreases with increasing temperature, while avalanche breakdown voltage increases with increasing temperature.
- A Zener diode can be used as a voltage regulator, a voltage reference, or a protection device in electronic circuits. It can maintain a constant voltage across its terminals, regardless of the variations in the input voltage or the load current, as long as the input voltage is higher than the breakdown voltage and the power dissipation of the diode is within its rating.



## Diode Application

A diode is a two-terminal electronic device that allows current to flow in only one direction. It has a low resistance in the forward direction and a high resistance in the reverse direction. Diodes are widely used in various fields of electronics for different purposes. Some of the common applications of diodes are:

- **Rectification:** A diode can convert an alternating current (AC) into a direct current (DC) by blocking the negative or positive half cycles of the AC waveform. This process is called rectification and it is the basic function of a diode. Rectifiers are used in power supplies, battery chargers, radio receivers, and many other devices that require a DC voltage .
- **Switching:** A diode can act as a switch that can turn on or off a circuit depending on the polarity of the input voltage. Diodes are used in digital logic circuits, memory devices, and pulse generators to control the flow of signals .
- **Source Isolation:** A diode can isolate a source from a load by preventing the reverse flow of current from the load to the source. This is useful for protecting the source from damage or interference caused by the load. Diodes are used in source isolation circuits, such as flyback diodes, free-wheeling diodes, and snubber diodes .
- **Voltage Reference:** A diode can provide a stable and precise voltage reference for various circuits by exploiting its forward voltage drop characteristic. The forward voltage drop of a diode depends on the current flowing through it and the temperature, but it is relatively constant for a given diode type. Diodes are used in voltage reference circuits, such as Zener diodes, Schottky diodes, and bandgap reference diodes .
- **Frequency Mixer:** A diode can mix two or more input signals of different frequencies and produce output signals of new frequencies that are the sum or difference of the input frequencies. This is useful for modulating or demodulating signals in communication systems. Diodes are used in frequency mixer circuits, such as ring modulators, balanced modulators, and product detectors .
- **Diode Detector:** A diode can detect the presence or absence of a signal by rectifying it and producing a DC output. This is useful for measuring the amplitude or envelope of a signal in radio receivers, audio amplifiers, and signal analyzers. Diodes are used in diode detector circuits, such as envelope detectors, peak detectors, and zero-crossing detectors .
- **Light Source:** A diode can emit light when a current flows through it in the forward direction. This is based on the phenomenon of electroluminescence, which occurs when electrons and holes recombine in a semiconductor material and release photons. Diodes are used as light sources, such as light-emitting diodes (LEDs), laser diodes, and organic light-emitting

diodes (OLEDs) .

- **Temperature and Light Sensor:** A diode can sense the temperature or light intensity by measuring its forward voltage drop or reverse current. The forward voltage drop of a diode decreases with increasing temperature, while the reverse current of a diode increases with increasing light intensity. Diodes are used as temperature and light sensors, such as thermistors, photodiodes, and solar cells .
- **Clipper and Clamper:** A diode can clip or clamp a signal to a desired voltage level by blocking or shunting the portions of the signal that exceed or fall below the voltage level. This is useful for limiting or shaping the signal waveform in various circuits. Diodes are used in clipper and clamper circuits, such as voltage limiters, voltage doublers, and voltage multipliers .
- **Circuit Protection:** A diode can protect a circuit from damage or malfunction caused by overvoltage, reverse polarity, or electrostatic discharge. Diodes are used in circuit protection circuits, such as transient voltage suppressors, diode bridges, and diode arrays .

### Diode Configuration

- A diode is an electrical device that allows current to flow in one direction only, blocking the reverse direction. It is a type of semiconductor device that consists of a p-n junction, where p and n are regions of opposite charge carriers (holes and electrons) in a crystal lattice.
- A diode has two terminals: an anode and a cathode. The anode is the positive terminal and the cathode is the negative terminal. The direction of the arrow in the diode symbol indicates the direction of conventional current flow when the diode is forward biased.
- A diode is forward biased when the anode is connected to a higher voltage than the cathode, and reverse biased when the anode is connected to a lower voltage than the cathode. When a diode is forward biased, it allows current to flow through it with a small voltage drop (usually 0.7 V for silicon diodes and 0.3 V for germanium diodes). When a diode is reverse biased, it blocks current flow and acts as an open circuit.
- A diode can be used for various applications, such as rectification, switching, clamping, voltage regulation, signal modulation, logic gates, etc. Some common types of diodes are Zener diodes, light-emitting diodes (LEDs), photodiodes, Schottky diodes, tunnel diodes, etc. Each type of diode has its own characteristics and configuration.
- A diode configuration is the arrangement of diodes in a circuit to perform a specific function. For example, a series diode configuration is when two or more diodes are connected in series, such that the current through them is the same. A parallel diode configuration is when two or more diodes are connected in parallel, such that the voltage across them is the same.

A bridge diode configuration is when four diodes are connected in a loop, such that they can convert an alternating current (AC) input into a direct current (DC) output.

## Half and Full Wave Rectification

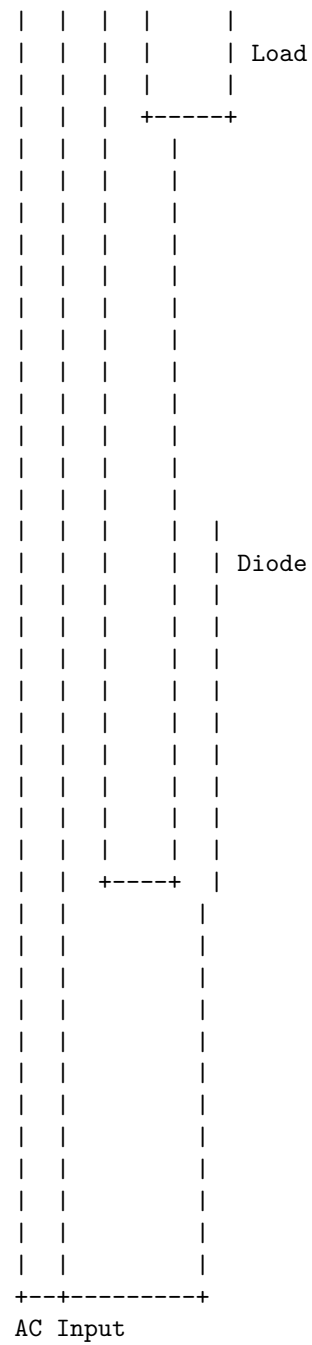
- Rectification is the process of converting an alternating current (AC) into a direct current (DC) by using one or more diodes.
- A diode is a semiconductor device that allows current to flow in one direction only. It has two terminals: anode and cathode.
- When the anode is positive with respect to the cathode, the diode is forward biased and conducts current. When the anode is negative with respect to the cathode, the diode is reverse biased and blocks current.
- There are two types of rectification: half wave and full wave.

## Half Wave Rectification

- A half wave rectifier uses a single diode to convert only one half cycle of the AC input into a pulsating DC output.
- The positive half cycle of the AC input passes through the diode and appears as the output. The negative half cycle of the AC input is blocked by the diode and does not appear as the output.
- The output frequency of a half wave rectifier is equal to the input frequency.
- The output voltage of a half wave rectifier is half of the peak input voltage.
- The output current of a half wave rectifier is proportional to the input voltage.
- The output power of a half wave rectifier is low and the ripple factor is high.
- The efficiency of a half wave rectifier is 40.6%.
- The circuit diagram of a half wave rectifier is shown below:

AC Input

+



### Full Wave Rectification

- A full wave rectifier uses two or four diodes to convert both half cycles of

the AC input into a pulsating DC output.

- The positive half cycle of the AC input passes through one pair of diodes and appears as the output. The negative half cycle of the AC input passes through another pair of diodes and is inverted to appear as the output.
- The output frequency of a full wave rectifier is twice the input frequency.
- The output voltage of a full wave rectifier is equal to the peak input voltage.
- The output current of a full wave rectifier is proportional to the input voltage.
- The output power of a full wave rectifier is high and the ripple factor is low.
- The efficiency of a full wave rectifier is 81.2%.
- There are two types of full wave rectifier: center-tapped and bridge.
- The circuit diagram of a center-tapped

### **Clippers for the notes of the Unit 1 - Semiconductor Diode in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING**

- A semiconductor diode is a diode made of semiconductor material, most often silicon. It is a device for controlling electrical currents so that they only flow the desired way.
- A diode has two terminals: an anode and a cathode. The anode is the positive terminal and the cathode is the negative terminal. The anode is connected to the p-type region of the semiconductor and the cathode is connected to the n-type region.
- A diode allows current to flow easily from the anode to the cathode, but severely restricts current from flowing in the opposite direction. This is because the p-n junction of the diode acts as a barrier for the reverse current.
- A diode is also known as a rectifier because it can change alternating current (AC) into pulsating direct current (DC). This is useful for many applications that require a constant voltage or current source.
- A clipper is a circuit that uses one or more diodes to clip or limit the voltage or current of a signal. A clipper can be used to protect a circuit from overvoltage, to remove unwanted parts of a signal, or to shape a signal into a desired form.
- A clipper can be classified into two types: series clipper and parallel clipper. A series clipper has the diode connected in series with the load, while a parallel clipper has the diode connected in parallel with the load.
- A clipper can also be classified into four types based on the polarity and level of clipping: positive clipper, negative clipper, biased positive clipper, and biased negative clipper. A positive clipper clips the positive half of the input signal, while a negative clipper clips the negative half. A biased clipper adds a DC voltage to the input signal before clipping it.
- A clipper can be designed using different components, such as resistors, capacitors, batteries, and Zener diodes. A Zener diode is a special type

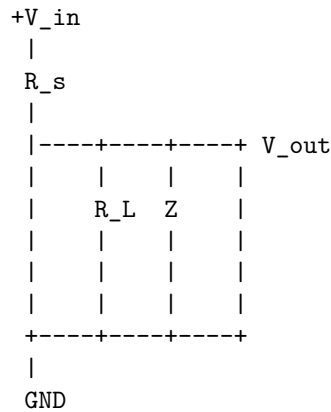
of diode that can conduct in both directions when the voltage across it exceeds a certain value, called the Zener voltage. A Zener diode can be used to clip a signal at a fixed voltage level.

### Clampers

- Clampers are electronic circuits that shift the dc level of the AC signal .
- Clampers are also known as DC voltage restorers or level shifter.
- Clampers are used to add the dc level to the ac input signal. The input swing of a waveform is equal to the output swing.
- Clampers can be classified as positive or negative, and biased or unbiased.
- A positive clamper circuit (negative peak clamper) outputs a purely positive waveform from an input signal; it offsets the input signal so that all of the waveform is greater than 0 V.
- A negative clamper circuit (positive peak clamper) outputs a purely negative waveform from an input signal; it offsets the input signal so that all of the waveform is less than 0 V.
- A biased clamper circuit adds a fixed dc voltage to the input signal, either positive or negative, depending on the polarity of the bias.
- An unbiased clamper circuit does not add any dc voltage to the input signal, but only shifts it up or down.
- An application of the clamper circuit is as a “DC restorer” in “composite video” circuitry in both television transmitters and receivers. An NTSC (US video standard) video signal “white level” corresponds to a minimum (12.5%) transmitted power.
- A clamper circuit consists of a diode, a capacitor, and a resistor. The diode conducts in one direction and blocks in the other direction. The capacitor charges and discharges according to the input signal. The resistor provides a discharge path for the capacitor.
- The operation of a clamper circuit depends on the polarity, bias, and frequency of the input signal, as well as the values of the diode, capacitor, and resistor. The output waveform can be analyzed using the concept of virtual ground.
- A clamper circuit can be simulated using software tools such as SPICE, Multisim, or MATLAB. The simulation can help to verify the theoretical analysis and design of the clamper circuit.

### Zener diode as shunt regulator

- A Zener diode is a special type of diode that can operate in the reverse breakdown region, where it maintains a nearly constant voltage across its terminals.
- A Zener diode can be used as a shunt voltage regulator, which is a device that regulates the output voltage across a load by diverting excess current to the ground .
- The basic circuit of a Zener diode shunt regulator is shown below:



- In this circuit,  $R_s$  is the series resistor,  $R_L$  is the load resistor,  $Z$  is the Zener diode,  $V_{in}$  is the input voltage, and  $V_{out}$  is the output voltage.
- The Zener diode is connected in parallel with the load, and it is reverse biased, meaning that the cathode is connected to the positive terminal and the anode is connected to the negative terminal.
- The Zener diode has a breakdown voltage,  $V_Z$ , which is the minimum reverse voltage required to make it conduct. When  $V_{in}$  is less than  $V_Z$ , the Zener diode does not conduct, and the output voltage is equal to the input voltage minus the voltage drop across  $R_s$ , which is given by Ohm's law:  $V_{Rs} = I_{Rs} * R_s$ , where  $I_{Rs}$  is the current through  $R_s$ .
- When  $V_{in}$  is greater than or equal to  $V_Z$ , the Zener diode starts to conduct, and the output voltage is equal to the breakdown voltage of the Zener diode,  $V_Z$ , regardless of the variations in  $V_{in}$  and  $R_L$ . This is because the excess current,  $I_Z$ , that flows through the Zener diode is equal to the difference between the current through  $R_s$  and the current through  $R_L$ :  $I_Z = I_{Rs} - I_{RL}$ , where  $I_{RL}$  is the current through  $R_L$ .
- The Zener diode shunt regulator has the following advantages:
  - It provides a better regulation over a wider range of load currents and input voltages.
  - It has a higher current capability, as the Zener diode can handle large currents in the breakdown region.
  - It is economical, as it is of low cost and requires few components.
- The Zener diode shunt regulator has the following disadvantages:
  - It has a poor efficiency, as the excess current is wasted as heat in the Zener diode and the series resistor.
  - It has a limited output voltage range, as the output voltage is fixed by the breakdown voltage of the Zener diode, which cannot be easily changed.
  - It has a poor load regulation, as the output voltage depends on the load resistance, which affects the current through the Zener diode.

## Voltage-Multiplier Circuits

- A voltage multiplier is an electronic circuit consisting of capacitors and diodes and is used to multiply or rise the voltage level of an AC signal.
- The voltage multiplier receives an AC voltage of lower value, converts it into a DC voltage and increases its voltage level.
- Voltage multipliers are classified as voltage doublers, triplers, quadruplers, etc, depending on the ratio of the output voltage to the input voltage.
- In theory, any desired amount of voltage multiplication can be obtained by cascading voltage doublers, but in practice, the output voltage is limited by the breakdown voltage of the diodes and capacitors, and the load current.
- Voltage multipliers can be used to generate a few volts for electronic appliances, to millions of volts for purposes such as high-energy physics experiments and lightning safety testing.

## Voltage Doubler Circuit

- A voltage doubler is a voltage multiplier circuit that produces an output voltage that is twice the peak input voltage.
- There are two types of voltage doublers: half-wave and full-wave.
- A half-wave voltage doubler consists of two capacitors and two diodes connected as shown below:

### Half-wave voltage doubler circuit

- The operation of the half-wave voltage doubler can be explained as follows:
  - During the positive half-cycle of the input AC voltage, diode D1 is forward biased and capacitor C1 is charged to the peak value of the input voltage  $V_p$ .
  - During the negative half-cycle of the input AC voltage, diode D2 is forward biased and capacitor C2 is charged to the peak value of the input voltage  $V_p$  through diode D1 and capacitor C1.
  - The output voltage across capacitor C2 is the sum of the voltages across C1 and C2, which is  $2V_p$ .
  - The output voltage is a pulsating DC voltage with a ripple frequency equal to the input frequency.
- A full-wave voltage doubler consists of four diodes and two capacitors connected as shown below:

### Full-wave voltage doubler circuit

- The operation of the full-wave voltage doubler can be explained as follows:
  - During the positive half-cycle of the input AC voltage, diode D1 is forward biased and capacitor C1 is charged to the peak value of the input voltage  $V_p$ .
  - During the negative half-cycle of the input AC voltage, diode D2 is forward biased and capacitor C2 is charged to the peak value of the input voltage  $V_p$  through diode D1 and capacitor C1.



- The output voltage across capacitor C2 is the sum of the voltages across C1 and C2, which is  $2V_p$ .
- The output voltage is a pulsating DC voltage with a ripple frequency equal to twice the input frequency.
- The full-wave voltage doubler has the advantage of higher output voltage, lower ripple, and better efficiency than the half-wave voltage doubler.

### Voltage Tripler Circuit

- A voltage tripler is a voltage multiplier circuit that produces an output voltage that is three times the peak input voltage.
- A voltage tripler circuit can be constructed by adding a capacitor and a diode to a half-wave voltage doubler circuit as shown below:

Voltage tripler circuit

- The operation of the voltage tripler circuit can be explained as follows:
  - During the positive half-cycle of the input AC voltage, diode D1 is forward biased and capacitor C1 is charged to the peak value of the input voltage  $V_p$ .
  - During the negative half-cycle of the input AC voltage, diode D2 is forward biased and capacitor C2 is charged to the peak value of the input voltage  $V_p$  through diode D1 and capacitor C1.
  - The voltage across capacitor C2 is  $2V_p$ .
  - During the next positive half-cycle of the input AC voltage, diode D3 is forward biased and capacitor C3 is charged to the peak value of the input voltage  $V_p$  through diode D2 and capacitor C2.
  - The voltage across capacitor C3 is  $3V_p$ .
  - The output voltage across capacitor C3 is the sum

### Special Purpose Two Terminal Devices

Two terminal devices are electronic components that have only two terminals or leads, such as resistors, capacitors, diodes, etc. They are usually passive devices that do not require external power to operate, but some of them can also be active devices that can amplify or switch signals, such as transistors, thyristors, etc.

Some of the two terminal devices have special properties or functions that make them useful for specific applications. These are called special purpose two terminal devices. Some examples of these devices are:

- **Zener diode:** A zener diode is a type of diode that can operate in the reverse breakdown region, where it exhibits a constant voltage drop across it regardless of the current flowing through it. This makes it useful for voltage regulation, reference, and protection circuits.

- **Light emitting diode (LED):** An LED is a type of diode that emits light when a forward current passes through it. The color and intensity of the light depend on the material and structure of the diode. LEDs are widely used for indicators, displays, lighting, and communication.
- **Photo diode:** A photo diode is a type of diode that generates a current when exposed to light. The amount of current depends on the intensity and wavelength of the light. Photo diodes are used for light detection, measurement, and conversion.
- **Varactor diode:** A varactor diode is a type of diode that has a variable capacitance depending on the reverse voltage applied to it. The capacitance decreases as the voltage increases. Varactor diodes are used for tuning, frequency modulation, and filtering circuits.
- **Tunnel diode:** A tunnel diode is a type of diode that has a very high conductivity in the forward direction due to quantum tunneling effect. It also has a negative resistance region, where the current decreases as the voltage increases. Tunnel diodes are used for high frequency oscillators, amplifiers, and switches.
- **Schottky diode:** A schottky diode is a type of diode that has a metal-semiconductor junction instead of a p-n junction. It has a lower forward voltage drop and a faster switching speed than a conventional diode. Schottky diodes are used for rectification, clamping, and logic circuits.
- **Silicon controlled rectifier (SCR):** An SCR is a type of thyristor that has four layers of alternating p and n type materials. It has three terminals: anode, cathode, and gate. It acts as a switch that can be turned on by a gate pulse and turned off by reducing the anode current. SCRs are used for power control, conversion, and protection.

### Light-Emitting Diodes

- A light-emitting diode (LED) is a semiconductor device that emits light when current flows through it.
- The light is produced by the recombination of electrons and electron holes in the semiconductor, a process called “electroluminescence” .
- The wavelength and color of the light depend on the energy band gap of the semiconductors used.
- LEDs are p-n junction devices made from extrinsic semiconductors. An n-type and a p-type semiconductor are put in contact with each other to form a p-n junction diode.
- When a forward voltage is applied across the p-n junction, electrons from the n-type region cross the depletion layer and recombine with holes from the p-type region, releasing photons in the process .
- The schematic symbol of an LED is shown below:

LED symbol

- LEDs have many advantages over conventional light sources, such as low power consumption, long lifespan, high efficiency, fast switching, and en-

vironmental friendliness .

- LEDs have many applications in various fields, such as displays, indicators, lighting, communication, sensing, and medical devices .

## Photo Diodes

- A photo diode is a type of semiconductor diode that converts light into electric current.
- It consists of a p-n junction or a p-i-n junction that is exposed to light through a transparent window or lens.
- When photons of sufficient energy strike the junction, they create electron-hole pairs that are separated by the electric field of the junction.
- The photo diode can operate in two modes: photovoltaic mode and photoconductive mode.
  - In photovoltaic mode, the photo diode is connected with zero or reverse bias, and the generated current is proportional to the light intensity. This mode is used for solar cells and power generation.
  - In photoconductive mode, the photo diode is connected with a reverse bias, and the generated current is proportional to the excess carriers created by the light. This mode is used for light detection and measurement.
- The main characteristics of a photo diode are:
  - Responsivity: the ratio of output current to input light power, measured in amperes per watt (A/W).
  - Quantum efficiency: the ratio of number of electrons generated to number of photons absorbed, measured in percentage (%).
  - Dark current: the leakage current that flows in the absence of light, measured in amperes (A).
  - Rise time: the time required for the output current to reach 90% of its final value after a step change in light intensity, measured in seconds (s).
  - Bandwidth: the frequency range over which the output current can vary without significant distortion, measured in hertz (Hz).
  - Spectral response: the variation of responsivity with wavelength of light, measured in A/W per nanometer (nm).
- The main applications of photo diodes are:
  - Optical communication: photo diodes are used to receive and convert optical signals into electrical signals in fiber optic networks and systems.
  - Optical sensing: photo diodes are used to measure light intensity, color, position, distance, speed, motion, etc. in various fields such as industrial, medical, environmental, etc.
  - Optical imaging: photo diodes are used to capture and process images

- in cameras, scanners, night vision devices, etc.
- Optical switching: photo diodes are used to control the transmission of light in optical switches, modulators, multiplexers, etc.

### Varactor Diodes

- A varactor diode is a type of diode that acts as a variable capacitor when reverse biased .
- The capacitance of a varactor diode depends on the applied voltage and the physical characteristics of the diode .
- The formula for the capacitance of a varactor diode is:

$$C = C_0 / (1 + V_r / V_0)^m$$

where  $C_0$  is the zero-bias capacitance,  $V_r$  is the reverse voltage,  $V_0$  is the built-in potential, and  $m$  is a constant that depends on the doping profile of the diode.

- Varactor diodes are widely used in applications that require voltage-controlled tuning, such as voltage-controlled oscillators, frequency multipliers, parametric amplifiers, and filters .
- Varactor diodes are also known as varicap diodes, tuning diodes, or variable reactance diodes .
- Varactor diodes are available in different packages and materials, such as silicon, gallium arsenide, surface mount, flip chip, ceramic, glass, and bare die .

### Tunnel Diodes

- A tunnel diode is a type of semiconductor diode that has effectively “negative resistance” due to the quantum mechanical effect called tunneling.
- Tunneling is the phenomenon where an electron can pass through a potential barrier that is higher than its kinetic energy.
- A tunnel diode is formed by a heavily doped p-n junction that is about 10 nm wide.
- The heavy doping results in a broken band gap, where conduction band electron states on the n-side are aligned with valence band hole states on the p-side.
- This allows electrons to tunnel from the n-side to the p-side with very little voltage applied.
- The tunnel diode was invented in August 1957 by Leo Esaki, Yuriko Kurose, and Takashi Suzuki when they were working at Tokyo Tsushin Kogyo, now known as Sony.
- The symbol of a tunnel diode is shown below:

Tunnel diode symbol

- The current-voltage (I-V) characteristic of a tunnel diode is shown below:

Tunnel diode I-V characteristic

- The I-V characteristic has three regions: forward bias, negative resistance, and reverse bias.
- In the forward bias region, the current increases rapidly as the voltage increases until it reaches a peak value (IP) at a low voltage (VP).
- In the negative resistance region, the current decreases as the voltage increases until it reaches a valley value (IV) at a higher voltage (VV).
- In the reverse bias region, the current increases again as the voltage increases, but at a much lower rate than in the forward bias region.
- The negative resistance region is useful for applications such as oscillators, amplifiers, and switches.
- Some advantages of tunnel diodes are: high speed, low noise, low power consumption, and simple circuitry.
- Some disadvantages of tunnel diodes are: low output voltage, low dynamic range, and temperature sensitivity.

## Unit 2 - Bipolar Junction Transistor

- A bipolar junction transistor (BJT) is a type of transistor that uses both electrons and holes as charge carriers.
- A transistor is a semiconductor device that can amplify or switch electrical signals.
- A BJT has three terminals: the emitter (E), the base (B), and the collector (C) .
- The emitter is the terminal that supplies the majority charge carriers (electrons for npn and holes for pnp) to the base.
- The base is the terminal that controls the flow of charge carriers from the emitter to the collector.
- The collector is the terminal that collects the charge carriers from the base and delivers them to the external circuit.
- A BJT can operate in three modes: cut-off, active, and saturation.
- In the cut-off mode, the base-emitter junction and the base-collector junction are both reverse biased, and no current flows through the transistor.
- In the active mode, the base-emitter junction is forward biased and the base-collector junction is reverse biased, and a small base current controls a large collector current.
- In the saturation mode, the base-emitter junction and the base-collector junction are both forward biased, and the collector current reaches its maximum value.
- A BJT can be used as a switch, an amplifier, or an oscillator, depending on the biasing and the circuit configuration.

## Transistor Construction

- A transistor is a three-layer semiconductor device that can act as an amplifier or a switch.
- A semiconductor is a material that can conduct electricity under certain conditions, such as when an electric field or a light is applied to it.
- The three layers of a transistor are made of either P-type or N-type semiconductor material, which have different electrical properties.
- P-type semiconductor material has a surplus of holes, which are the absence of electrons. Holes can move through the material when an electric field is applied, creating a positive current.
- N-type semiconductor material has a surplus of electrons, which can move through the material when an electric field is applied, creating a negative current.
- The three layers of a transistor are arranged in either NPN or PNP configuration, as shown in the figure below.

### Transistor configuration

- The middle layer is called the base, and the outer layers are called the emitter and the collector.
- The base is very thin and lightly doped, meaning it has fewer holes or electrons than the emitter and the collector, which are heavily doped.
- The emitter and the collector are connected to an external circuit, and the base is connected to a control voltage or current.
- The transistor works by controlling the flow of current from the emitter to the collector, depending on the voltage or current applied to the base.
- In an NPN transistor, the emitter is N-type and the collector is P-type. When a positive voltage is applied to the base, it attracts electrons from the emitter, which then flow through the base to the collector, creating a large output current. This is called the forward active mode of the transistor, and it works as an amplifier.
- When a negative voltage is applied to the base, it repels electrons from the emitter, and no current flows from the emitter to the collector. This is called the cutoff mode of the transistor, and it works as a switch.
- In a PNP transistor, the emitter is P-type and the collector is N-type. The operation is similar to the NPN transistor, but with opposite polarities. A negative voltage at the base causes a large output current from the emitter to the collector, and a positive voltage at the base causes no output current.

## Operation of Bipolar Junction Transistor

- A bipolar junction transistor (BJT) is a type of transistor that uses both electron and hole charge carriers. In contrast, unipolar transistors, such as field-effect transistors, only use one kind of charge carrier.
- A BJT has three terminals: the base (B), the collector (C), and the emit-

ter (E). The base is the control terminal, the collector is the output terminal, and the emitter is the input terminal. The base-emitter junction is forward-biased, while the base-collector junction is reverse-biased.

- A BJT can operate in three regions: the active region, the saturation region, and the cutoff region. The region of operation depends on the biasing voltages and currents applied to the terminals.
- In the active region, the BJT acts as an amplifier. The base current controls the collector current, which is proportional to the base-emitter voltage. The current gain, or beta, is the ratio of the collector current to the base current. The voltage gain is the ratio of the output voltage to the input voltage.
- In the saturation region, the BJT acts as a switch that is fully on. The collector current is equal to the saturation current, which is the maximum current that can flow through the collector-emitter circuit. The collector-emitter voltage is very low, close to zero. The base current is higher than the minimum required to saturate the transistor.
- In the cutoff region, the BJT acts as a switch that is fully off. The collector current is zero, and the collector-emitter voltage is equal to the supply voltage. The base current is also zero, or very low.
- A BJT can be either npn or pnp, depending on the type and arrangement of the semiconductor layers. An npn transistor has a thin layer of p-type material sandwiched between two layers of n-type material. A pnp transistor has a thin layer of n-type material sandwiched between two layers of p-type material. The operation of the two types of transistors is similar, except that the polarity of the voltages and currents are reversed.

### **Amplification action for the notes of the Unit 2 - Bipolar Junction Transistor in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING**

- A bipolar junction transistor (BJT) is a three-terminal device that can amplify the flow of current by controlling the current in one terminal (base) with the voltage applied at another terminal (emitter) .
- A BJT has two types of terminals: N-type and P-type, which are formed by doping a semiconductor material with impurities. A BJT can have either NPN or PNP configuration, depending on the arrangement of the terminals .
- A BJT can operate in three regions: cut-off, active, and saturation. In the cut-off region, the base-emitter junction is reverse biased and the base-collector junction is either reverse or forward biased, resulting in no current flow in the collector. In the saturation region, both the base-emitter and base-collector junctions are forward biased, resulting in maximum current flow in the collector. In the active region, the base-emitter junction is forward biased and the base-collector junction is reverse biased, resulting in a controlled current flow in the collector .

- A BJT amplifier is a circuit that uses a BJT in the active region to increase the strength of the input signal and produce an amplified output. The input signal is applied to the base terminal and the output signal is taken from the collector terminal. The emitter terminal is usually connected to a common ground .
- Depending on the quantity amplified by the circuit, the BJT amplifier can be a voltage, current, or power amplifier. A voltage amplifier increases the voltage of the input signal and has a high input impedance and a low output impedance. A current amplifier increases the current of the input signal and has a low input impedance and a high output impedance. A power amplifier increases the power of the input signal and has a moderate input and output impedance .
- A BJT amplifier can also be classified based on the configuration of the terminals: common emitter, common base, or common collector. A common emitter amplifier has the emitter terminal as the common terminal for both input and output. It has a high voltage gain, a moderate current gain, and a moderate power gain. A common base amplifier has the base terminal as the common terminal for both input and output. It has a low voltage gain, a high current gain, and a high power gain. A common collector amplifier has the collector terminal as the common terminal for both input and output. It has a low voltage gain, a low current gain, and a low power gain .
- A BJT amplifier can be used for various applications, such as audio amplification, radio frequency amplification, signal processing, switching, and logic circuits .

### **Common Base Configuration of BJT**

- The common base configuration is one of the three basic ways to connect a bipolar junction transistor (BJT) as an amplifier.
- In this configuration, the base terminal of the transistor is a common terminal to both the input and output signals, hence its name common base (CB) .
- The input signal is applied between the emitter and the base, and the output signal is taken from the collector and the base .
- The common base configuration has the following characteristics :
  - High voltage gain
  - Low current gain
  - High input impedance
  - Low output impedance
  - No phase inversion between input and output
  - High frequency response
- The common base configuration is less common as an amplifier than compared to the more popular common emitter (CE) or common collector (CC) configurations .
- The common base configuration is mainly used for impedance matching,



current buffering, and high frequency applications .

### **Common Emitter**

- A common emitter (CE) amplifier is one of the three basic single-stage bipolar junction transistor (BJT) amplifier topologies, typically used as a voltage amplifier .
- A BJT is a type of transistor that uses both electrons and holes as charge carriers.
- In a CE amplifier, the emitter terminal of the BJT is common to both the input and output circuits .
- The input signal is applied at the base terminal and the output signal is obtained at the collector terminal.
- The base current controls the collector current, and the ratio of change in collector current to change in base current is defined as the current gain of the CE amplifier.
- The emitter current is the sum of the base current and the collector current.
- The CE amplifier offers high current gain (typically 200), medium input resistance and high output resistance.
- The CE amplifier also has a high voltage gain, which is the ratio of change in output voltage to change in input voltage.
- The CE amplifier requires proper biasing to operate in the active region, where the transistor acts as a linear amplifier.
- The CE amplifier can be analyzed using the hybrid-pi model, which is a small-signal model that approximates the BJT behavior using resistors, capacitors and controlled sources.
- The CE amplifier can be used for various applications, such as audio amplification, signal processing, oscillators, etc.

### **Common Collector Configuration**

- In this configuration, the base terminal of the transistor serves as the input, the emitter terminal is the output and the collector terminal is common for both input and output.
- The collector terminal is grounded so the common collector configuration is also known as grounded collector configuration.
- Sometimes common collector configuration is also referred to as emitter follower, voltage follower, common collector amplifier, CC amplifier, or CC configuration .
- The common collector amplifier is often used as a voltage buffer, because it has a high input impedance, a low output impedance and a non-inverting voltage gain of approximately one .
- The input circuit is connected between emitter and base and the output is taken from the collector and emitter.
- The load resistor in the common-collector amplifier circuit receives both

the base and collector currents, being placed in series with the emitter.

- The common collector amplifier has a high current gain, a high power gain, but a low voltage gain.
- The common collector amplifier has a high input resistance and a low output resistance, which makes it suitable for impedance matching.
- The common collector amplifier has a high stability against temperature variations and a good frequency response.

### Unit 3 - Field Effect Transistor

- A field effect transistor (FET) is a type of transistor that uses an electric field to control the flow of current in a semiconductor.
- FETs have three terminals: source, gate, and drain. The source is where the current enters the device, the gate is where the electric field is applied, and the drain is where the current leaves the device.
- There are two main types of FETs: junction FETs (JFETs) and metal-oxide-semiconductor FETs (MOSFETs). JFETs use a pn-junction as the gate, while MOSFETs use a metal-oxide layer as the gate .
- JFETs can be either n-channel or p-channel, depending on the type of semiconductor used. N-channel JFETs have a higher current and lower voltage than p-channel JFETs.
- MOSFETs can also be either n-channel or p-channel, as well as enhancement-mode or depletion-mode, depending on the polarity and operation of the gate voltage. Enhancement-mode MOSFETs are normally off and require a positive gate voltage to turn on, while depletion-mode MOSFETs are normally on and require a negative gate voltage to turn off.
- FETs are widely used in electronic circuits, such as amplifiers, switches, sensors, and logic gates, because they have high input impedance, low noise, and high speed.

#### Construction and Characteristic of JFETs

- A JFET (Junction Field Effect Transistor) is a three-terminal semiconductor device that can be used as a voltage-controlled resistor, switch or amplifier .
- A JFET is constructed using a long channel of semiconductor material, either N-type or P-type, with two ohmic contacts at the ends, called the source (S) and the drain (D), and a reverse-biased PN junction on each side, called the gate (G) .
- The symbols and basic construction for both N-channel and P-channel JFETs are shown below:

JFET symbols and construction

- The current flow in a JFET is controlled by the voltage applied between the gate and the source,  $V_{GS}$ , which modulates the width of the channel

and thus the resistance of the device .

- The JFET has two operating regions: the ohmic region and the saturation region .
- In the ohmic region, the JFET behaves like a linear resistor, with a low drain-source voltage,  $V_{DS}$ , and a high channel width .
- In the saturation region, the JFET behaves like a constant current source, with a high  $V_{DS}$  and a low channel width .
- The transition from the ohmic region to the saturation region occurs when  $V_{DS}$  reaches a value called the pinch-off voltage,  $V_P$ , which depends on the doping level and the geometry of the device .
- The characteristic curves of a JFET show the relationship between the drain current,  $I_D$ , and the drain-source voltage,  $V_{DS}$ , for different values of  $V_{GS}$  .
- The characteristic curves of an N-channel JFET are shown below:

JFET characteristic curves

- The transfer characteristic of a JFET shows the relationship between the drain current,  $I_D$ , and the gate-source voltage,  $V_{GS}$ , for a constant value of  $V_{DS}$  .
- The transfer characteristic of an N-channel JFET is shown below:

JFET transfer characteristic

- The transfer characteristic of a JFET is nonlinear and can be approximated by a quadratic equation of the form:  $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ , where  $I_{DSS}$  is the maximum drain current when  $V_{GS} = 0$  and  $V_P$  is the pinch-off voltage .
- The JFET has some advantages over the BJT, such as high input impedance, low noise, low power consumption and high temperature stability .
- The JFET also has some disadvantages, such as low gain, limited frequency response and susceptibility to damage by static electricity .

### Transfer Characteristic of FET

- The transfer characteristic of a field-effect transistor (FET) is a curve that shows the relationship between the input voltage ( $V_{GS}$ ) and the output current ( $I_D$ ) for a given value of the drain-source voltage ( $V_{DS}$ ).
- The transfer characteristic can be derived from the drain characteristic, which is a set of curves that show the variation of  $I_D$  with  $V_{DS}$  for different values of  $V_{GS}$ .
- The transfer characteristic can be used to determine the operating point, the transconductance, and the amplification factor of the FET.
- The transfer characteristic can be expressed by a mathematical equation, such as the Shockley equation for the junction FET (JFET) or the square-law equation for the metal-oxide-semiconductor FET (MOSFET).

- The transfer characteristic can be plotted on a linear or a logarithmic scale, depending on the type and the region of operation of the FET.
- The transfer characteristic can be normalized by using the ratios  $I_D/I_{DSS}$  and  $V_{GS}/V_P$ , where  $I_{DSS}$  is the saturation current and  $V_P$  is the pinch-off voltage of the FET. This results in a universal transfer characteristic that can be applied to any FET with the same geometry and material.
- The transfer characteristic can be affected by the temperature, the channel length modulation, and the body effect of the FET.

: Universal Transfer Characteristics for FET | Circuit Analysis. (n.d.). Retrieved March 15, 2023, from <https://www.eeeguide.com/universal-transfer-characteristics-for-fet/>

### **MOSFET (MOS) (Depletion and Enhancement) Type**

- MOSFET stands for Metal-Oxide-Semiconductor Field-Effect Transistor. It is a type of FET that uses an electric field to control the conductivity of a channel between the source and drain terminals.
- MOSFETs can be classified into two major types: depletion mode and enhancement mode, depending on whether the channel is present or absent at zero gate-source voltage.
- Depletion mode MOSFETs have a channel that is fabricated during manufacturing. They conduct current between the source and drain terminals even when the gate-source voltage is zero. Applying a reverse voltage to the gate reduces the channel width and the current flow. Applying a forward voltage increases the channel width and the current flow.
- Enhancement mode MOSFETs have no channel during manufacturing. They do not conduct current when the gate-source voltage is zero. Applying a forward voltage to the gate creates a channel and allows current to flow. Applying a reverse voltage does not affect the channel or the current flow.
- Both depletion and enhancement mode MOSFETs can be further divided into n-channel and p-channel types, depending on whether the channel is made of n-type or p-type semiconductor material. For n-channel MOSFETs, the substrate is p-type and the gate voltage is positive with respect to the source. For p-channel MOSFETs, the substrate is n-type and the gate voltage is negative with respect to the source.
- MOSFETs have many applications in digital and analog circuits, such as switches, amplifiers, converters, inverters, oscillators, and logic gates. They have advantages over other types of transistors, such as high input impedance, low power consumption, high switching speed, and compatibility with integrated circuits.

### **Transfer Characteristic of Field Effect Transistor**

- The transfer characteristic of a field effect transistor (FET) is the curve that shows the relation between the gate voltage and the drain current,

while keeping the drain-source voltage constant .

- The transfer characteristic can be used to determine the transconductance of the FET, which is a measure of the gain or amplification that the device can provide.
- The transfer characteristic can be obtained experimentally by applying a fixed drain-source voltage and varying the gate-source voltage, and measuring the corresponding drain current.
- The shape of the transfer characteristic depends on the type and mode of the FET. For example, a junction field effect transistor (JFET) can operate in either enhancement mode or depletion mode, and the transfer characteristic will be different for each mode.
- The transfer characteristic can also be affected by the temperature, the channel length, and the channel width of the FET.
- The transfer characteristic can be represented by a mathematical equation that relates the gate voltage and the drain current, using some parameters such as the pinch-off voltage, the threshold voltage, and the saturation current.
- The transfer characteristic can be plotted on a graph, where the horizontal axis is the gate voltage and the vertical axis is the drain current. The graph can show the linear region, the saturation region, and the cut-off region of the FET operation .
- The transfer characteristic can be used to analyze the performance and the behavior of the FET in different circuits and applications.

## Unit 4 - Operational Amplifiers

- An operational amplifier (op amp) is an analog circuit block that takes a differential voltage input and produces a single-ended voltage output.
- Op amps usually have three terminals: two high-impedance inputs and a low-impedance output port. The inverting input is denoted with a minus (-) sign, and the non-inverting input uses a positive (+) sign .
- Op amps can be classified into four types based on their input and output characteristics:
  - Voltage amplifiers take voltage in and produce a voltage at the output.
  - Current amplifiers receive a current input and produce a current output.
  - Transconductance amplifiers convert a voltage input to a current output.
  - Transresistance amplifiers convert a current input to a voltage output.
- Op amps can be used for various applications, such as signal conditioning, filtering, amplification, mixing, oscillation, integration, differentiation, etc .
- Op amps have some ideal characteristics, such as infinite input impedance, zero output impedance, infinite voltage gain, infinite bandwidth, zero

noise, etc .

- However, in reality, op amps have some limitations and non-idealities, such as finite input impedance, non-zero output impedance, finite voltage gain, finite bandwidth, noise, offset, distortion, etc .
- These non-idealities can affect the performance and accuracy of op amps and their applications, and therefore need to be considered and compensated for in the design and analysis of op amp circuits .

#### **Introduction for the notes of the Unit 4 - Operational Amplifiers in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING**

- An operational amplifier (op-amp) is a high-gain, direct-coupled, differential-input amplifier that can perform various linear and nonlinear operations on input signals.
- The term “operational” refers to the fact that op-amps were originally designed to perform mathematical operations, such as addition, subtraction, integration, and differentiation, in analog computers.
- The basic symbol of an op-amp is shown below:

op-amp symbol

- The op-amp has two input terminals: the inverting input (-) and the non-inverting input (+), and one output terminal. It also has two power supply terminals: the positive supply (V+) and the negative supply (V-).
- The op-amp amplifies the difference between the input voltages ( $V_+ - V_-$ ) and produces an output voltage ( $V_{out}$ ) that is proportional to the input difference.
- The op-amp has a very high input impedance ( $Z_{in}$ ), which means that it draws very little current from the input sources. It also has a very low output impedance ( $Z_{out}$ ), which means that it can drive a large current to the output load.
- The op-amp has a very high open-loop gain ( $A$ ), which is the ratio of the output voltage to the input difference without any feedback. The open-loop gain is typically in the order of  $10^5$  to  $10^6$ , and it varies with frequency and temperature.
- The op-amp is usually connected with a feedback network, which is a circuit that connects the output to the input. The feedback network can control the gain, bandwidth, stability, and linearity of the op-amp. The feedback network can be either negative or positive, depending on whether it subtracts or adds the output to the input.
- The op-amp can be used to implement various linear and nonlinear circuits, such as amplifiers, filters, oscillators, comparators, integrators, differentiators, etc. The op-amp is a versatile and widely used component in electronic systems.

## Op-Amp Basics

- An operational amplifier (op-amp) is a device that can amplify the difference between two input voltages and produce a single-ended output voltage .
- An op-amp has two input terminals, labeled as  $V+$  (non-inverting input) and  $V-$  (inverting input), and one output terminal, labeled as  $V_{out}$  .
- An op-amp also has two power supply terminals, labeled as  $V_{s+}$  (positive supply) and  $V_{s-}$  (negative supply), which provide the maximum and minimum output voltages respectively .
- An op-amp can be considered as a differential-to-single-ended converter, as it converts a differential input signal ( $V+ - V-$ ) to a single-ended output signal ( $V_{out}$ ) .
- An op-amp has a very high open-loop gain, which means that the output voltage is proportional to the input voltage difference multiplied by a large constant .
- An op-amp has a very high input impedance, which means that it draws very little current from the input sources .
- An op-amp has a very low output impedance, which means that it can drive a large current to the load without significant voltage drop .
- An op-amp exhibits common-mode rejection, which means that it ignores any voltage component that is present in both input signals, such as noise or DC offset .
- An op-amp can be used in various configurations, such as inverting amplifier, non-inverting amplifier, summing amplifier, difference amplifier, integrator, differentiator, etc., by connecting external resistors and capacitors .

## Practical Op-Amp Circuits

An operational amplifier (op-amp) is a versatile device that can be used to amplify signals, filter signals, perform mathematical operations and more. Op-amps are usually used in conjunction with passive components such as resistors and capacitors to create various circuits with different functions. Some of the most common and fundamental op-amp circuits are:

- **Voltage follower:** This is the simplest op-amp circuit, where the output voltage is equal to the input voltage. It does not require any external components and it provides a high input impedance and a low output impedance. It can be used to isolate or buffer a signal from a source or a load.
- **Inverting op-amp:** This circuit produces an output voltage that is proportional to the input voltage, but with an opposite sign. It requires a feedback resistor ( $R_2$ ) and an input resistor ( $R_1$ ) to set the gain of the circuit, which is equal to  $-R_2/R_1$ . It can be used to invert or amplify a signal.

- **Non-inverting op-amp:** This circuit produces an output voltage that is proportional to the input voltage, but with the same sign. It requires a feedback resistor ( $R_2$ ) and a resistor ( $R_1$ ) connected to the ground to set the gain of the circuit, which is equal to  $1 + R_2/R_1$ . It can be used to amplify a signal without inverting it.
- **Non-inverting summing amplifier:** This circuit produces an output voltage that is proportional to the sum of two or more input voltages, but with the same sign. It requires a feedback resistor ( $R_f$ ) and a resistor ( $R_1$ ) for each input voltage to set the gain of the circuit, which is equal to  $1 + R_f/R_1$ . It can be used to add or mix signals.
- **Inverting summing amplifier:** This circuit produces an output voltage that is proportional to the sum of two or more input voltages, but with an opposite sign. It requires a feedback resistor ( $R_f$ ) and a resistor ( $R_1$ ) for each input voltage to set the gain of the circuit, which is equal to  $-R_f/R_1$ . It can be used to subtract or mix signals.
- **Differential amplifier:** This circuit produces an output voltage that is proportional to the difference of two input voltages. It requires four resistors ( $R_1$ ,  $R_2$ ,  $R_3$  and  $R_4$ ) to set the gain of the circuit, which is equal to  $R_2/R_1$  if  $R_3 = R_1$  and  $R_4 = R_2$ . It can be used to measure or amplify a differential signal.
- **Integrator:** This circuit produces an output voltage that is proportional to the integral of the input voltage over time. It requires a feedback capacitor ( $C_f$ ) and an input resistor ( $R_1$ ) to set the time constant of the circuit, which is equal to  $R_1 C_f$ . It can be used to perform integration or low-pass filtering.
- **Differentiator:** This circuit produces an output voltage that is proportional to the derivative of the input voltage over time. It requires a feedback resistor ( $R_f$ ) and an input capacitor ( $C_1$ ) to set the time constant of the circuit, which is equal to  $R_f C_1$ . It can be used to perform differentiation or high-pass filtering.
- **Comparator:** This circuit produces an output voltage that is either high or low, depending on whether the input voltage is higher or lower than a reference voltage. It does not require any feedback components and it operates in an open-loop mode. It can be used to compare or detect signals.
- **Oscillator:** This circuit produces an output voltage that oscillates at a certain frequency and amplitude. It requires a feedback network that provides a phase shift of 180 degrees and a gain greater than 1 at the desired frequency. It can be used to generate or modulate signals.

These are some of the basic and practical op-amp circuits that can be used for various applications in electronics engineering. However, there are many more



op-amp circuits that can be designed and analyzed using the same principles and techniques.

### Inverting Amplifier

- An inverting amplifier is a type of operational amplifier circuit that produces an output signal that is 180 degrees out of phase with the input signal .
- This means that if the input signal is positive, the output signal will be negative, and vice versa .
- An inverting amplifier can be used to amplify or invert a signal, depending on the values of the resistors in the circuit .
- The basic circuit diagram of an inverting amplifier is shown below:

Inverting amplifier circuit diagram

- The circuit consists of an operational amplifier, an input resistor  $R_i$ , and a feedback resistor  $R_f$ .
- The input signal  $V_i$  is applied to the inverting input terminal (-) of the op-amp, while the non-inverting input terminal (+) is connected to the ground.
- The output signal  $V_o$  is taken from the output terminal of the op-amp.
- The voltage gain of the inverting amplifier is given by the formula:

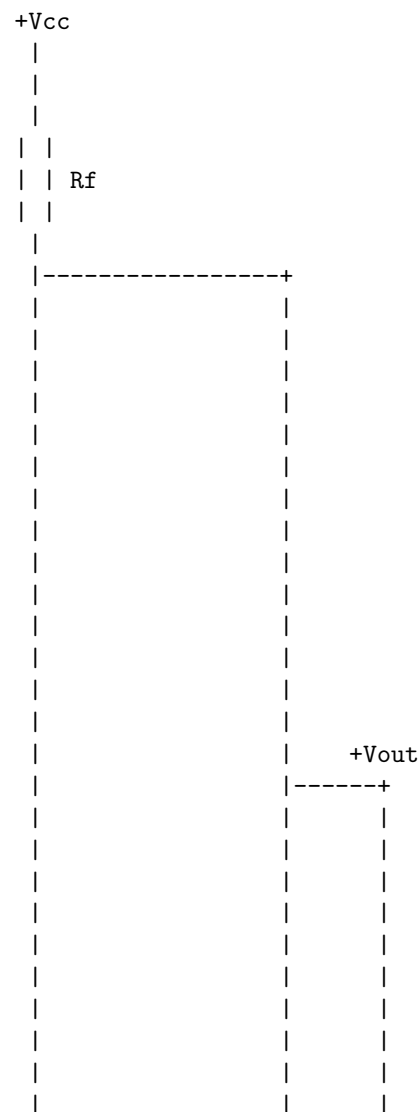
Inverting amplifier gain formula

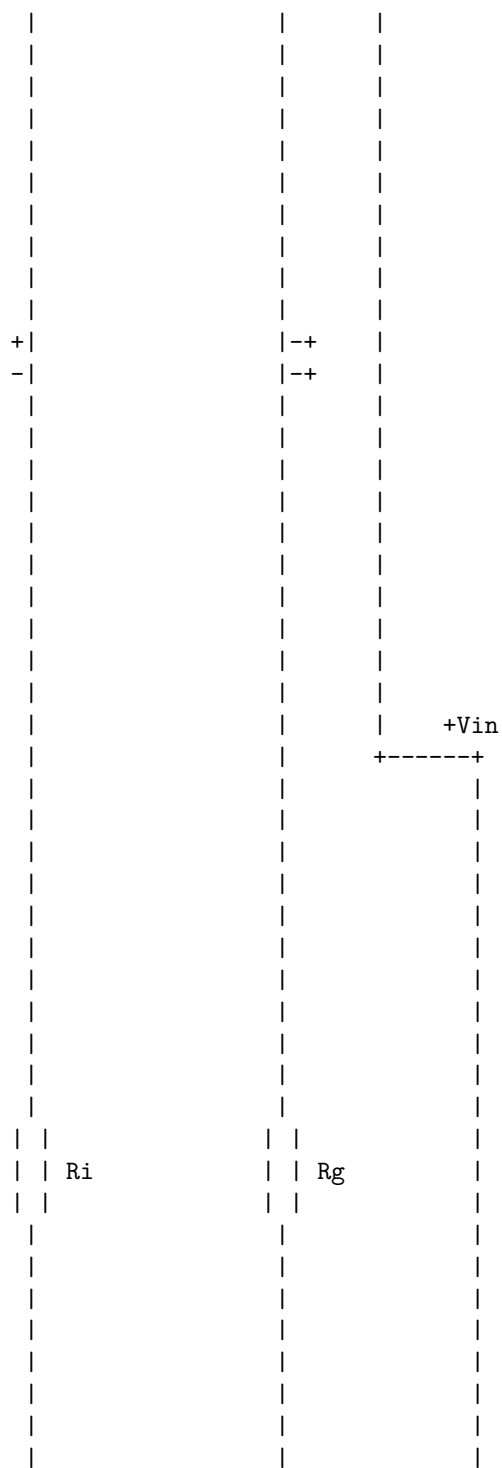
- The gain is negative, indicating that the output signal is inverted with respect to the input signal .
- The gain can be adjusted by changing the values of  $R_i$  and  $R_f$ . A larger  $R_f$  or a smaller  $R_i$  will result in a higher gain, and vice versa .
- The input impedance of the inverting amplifier is equal to the value of  $R_i$ , while the output impedance is very low .
- Some applications of the inverting amplifier are:
  - Signal inversion: The inverting amplifier can be used to reverse the polarity of a signal, such as converting a positive voltage to a negative voltage, or a sine wave to a cosine wave .
  - Signal amplification: The inverting amplifier can be used to increase the amplitude of a signal, such as boosting a weak audio signal or a sensor signal .
  - Signal subtraction: The inverting amplifier can be used to subtract two signals, such as removing a DC offset or a common-mode noise from a differential signal.
  - Signal mixing: The inverting amplifier can be used to combine multiple signals, such as adding different audio signals or waveforms.
  - Negative feedback: The inverting amplifier can be used to provide negative feedback to a system, such as stabilizing the output voltage

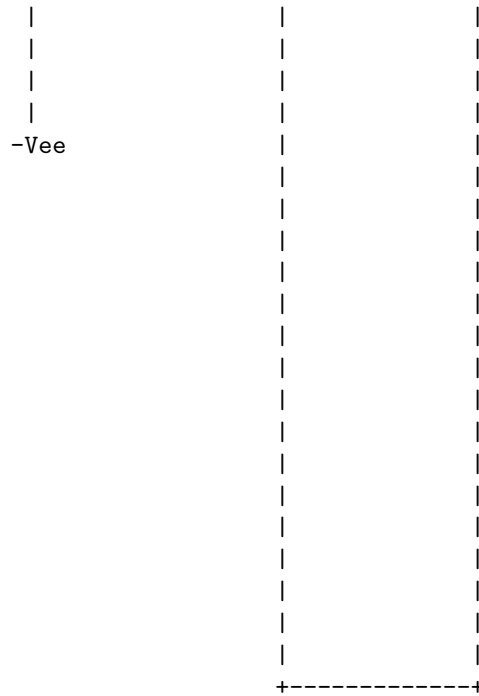
of a power supply or the frequency of an oscillator.

### Non-inverting Amplifier

- A non-inverting amplifier is an op-amp circuit configuration that produces an amplified output signal and this output signal of the non-inverting op-amp is in-phase with the applied input signal .
- In other words, a non-inverting amplifier behaves like a voltage follower circuit.
- The basic circuit diagram of a non-inverting amplifier is shown below:







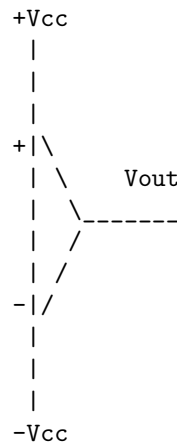
- The input signal is applied to the non-inverting (+) input terminal of the op-amp and the output signal is taken from the output terminal of the op-amp .
- The feedback resistor  $R_f$  is connected between the output and the inverting (-) input terminal of the op-amp .
- The input resistor  $R_i$  is connected between the inverting input terminal and the ground .
- The resistor  $R_g$  is connected between the non-inverting input terminal and the ground to balance the input bias currents .
- The voltage gain of the non-inverting amplifier is given by the formula :

$$A_v = 1 + (R_f / R_i)$$

- Where  $A_v$  is the voltage gain,  $R_f$  is the feedback resistor, and  $R_i$  is the input resistor.
- The voltage gain of the non-inverting amplifier is always greater than or equal to 1 .
- The input impedance of the non-inverting amplifier is very high, which means that it draws very little current from the input source .
- The output impedance of the non-inverting amplifier is very low

### Unit Follower for the notes of the Unit 4 - Operational Amplifiers in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING

- A unit follower, also known as a voltage follower, buffer, or unity-gain amplifier, is a simple op-amp circuit that produces an output voltage equal to the input voltage .
- A unit follower is created by directly connecting the output of the op-amp to the inverting (-) input, and applying the input voltage to the non-inverting (+) input .
- The circuit diagram of a unit follower is shown below:



- The working principle of a unit follower is based on the negative feedback and the virtual short concept of an ideal op-amp .
- The negative feedback forces the output voltage to be equal to the input voltage, so that the voltage difference between the two inputs is zero .
- The virtual short concept implies that the inverting and non-inverting inputs are at the same potential, so the input voltage is also equal to the output voltage .
- The advantages of a unit follower are :
  - It has a very high input impedance and a very low output impedance, which makes it ideal for isolating circuits from each other and preventing loading effects.
  - It has a unity voltage gain, which means it does not alter the amplitude or phase of the input signal.
  - It has a very wide bandwidth, which means it can pass a wide range of frequencies without distortion.
  - It has a very low noise and distortion, which means it preserves the quality of the input signal.
- The applications of a unit follower are :
  - It can be used as a buffer between different stages of a circuit, such as filters, amplifiers, or sensors, to avoid signal loss or interference.

- It can be used as a level shifter, to shift the DC level of a signal without changing its AC component.
- It can be used as a voltage regulator, to provide a stable and constant output voltage from a variable input voltage source.
- It can be used as a current booster, to increase the current driving capability of a circuit without changing its voltage level.

### **Summing Amplifier**

- A summing amplifier is an op amp circuit that can combine numbers of input signals to a single output that is the weighted sum of the applied inputs .
- The summing amplifier is one variation of inverting amplifier. In inverting amplifier there is only one voltage signal applied to the inverting input as shown below :

Inverting amplifier

- The output voltage of the inverting amplifier is given by:

Output voltage of inverting amplifier

- In summing amplifier, there are two or more voltage signals applied to the inverting input as shown below :

Summing amplifier

- The output voltage of the summing amplifier is given by:

Output voltage of summing amplifier

- The summing amplifier can be used to perform arithmetic operations such as addition, subtraction, scaling, and averaging of the input signals .
- The summing amplifier can also be used to mix audio signals, convert digital signals to analog signals, and implement feedback control systems .
- The summing amplifier is a very flexible circuit indeed, enabling us to effectively “Add” or “Sum” (hence its name) together several individual input signals.

### **Integrator**

- An integrator is an electronic circuit that performs the mathematical operation of integration with respect to time. It is based on an operational amplifier (op-amp) and a feedback capacitor.
- The basic configuration of an op-amp integrator is shown below:

Op-amp integrator circuit

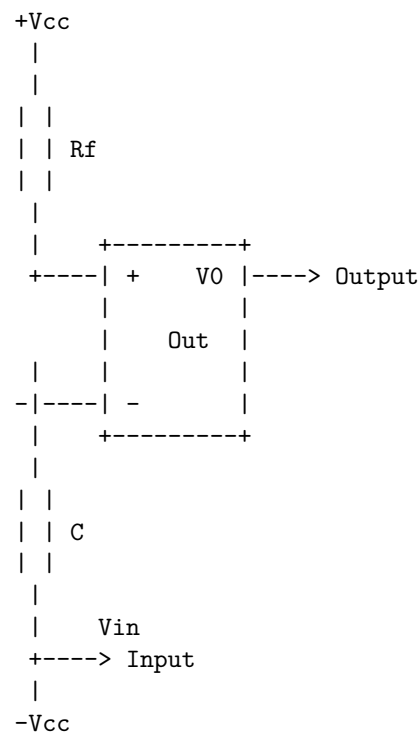
- The input voltage  $V_{in}$  is applied to the inverting terminal of the op-amp through a resistor  $R$ . The output voltage  $V_{out}$  is fed back to the same terminal through a capacitor  $C$ . The non-inverting terminal is grounded.
- The output voltage  $V_{out}$  is given by the following equation:

Op-amp integrator output equation

- The output voltage is proportional to the integral of the input voltage with respect to time. The constant of integration is determined by the initial condition of the capacitor.
- The integrator can be used to perform various functions, such as generating waveforms, filtering signals, measuring areas, and converting signals from analog to digital.

### Differentiator

- A differentiator is a circuit that performs the mathematical operation of differentiation, that is, it produces an output voltage that is proportional to the rate of change of the input voltage.
- A differentiator can be constructed using an operational amplifier (op-amp) with a capacitor and a resistor in the feedback loop.
- The basic circuit diagram of a differentiator is shown below:



- The input voltage is applied to the inverting terminal of the op-amp

through the capacitor C, while the non-inverting terminal is grounded. The resistor Rf provides negative feedback to the op-amp.

- The output voltage V0 is given by the formula:

$$V_0 = -R_f * dV_{in}/dt$$

- Where  $dV_{in}/dt$  is the derivative of the input voltage with respect to time.
- The negative sign indicates that the output voltage is inverted with respect to the input voltage.
- The capacitor C blocks any DC component of the input voltage and allows only the AC component to pass through. The resistor Rf determines the gain of the circuit and prevents the output from saturating.
- The differentiator circuit can be used to perform various functions such as edge detection, waveform generation, frequency modulation, etc. in analog circuits .

### Differential and Common-Mode Operation

- A differential amplifier is a circuit that can accept two input signals and amplify the difference between these two input signals.
- An op-amp is a type of differential amplifier that has a very high voltage gain and a very high input impedance.
- The differential input voltage of an op-amp is the voltage difference between the two input terminals, denoted by  $V_d = V_+ - V_-$ .
- The common-mode input voltage of an op-amp is the average of the two input voltages, denoted by  $V_{cm} = (V_+ + V_-)/2$ .
- The differential output voltage of an op-amp is the voltage difference between the two output terminals, denoted by  $V_o = V_{o+} - V_{o-}$ .
- The common-mode output voltage of an op-amp is the average of the two output voltages, denoted by  $V_{ocom} = (V_{o+} + V_{o-})/2$ .
- The differential mode of operation of an op-amp is when the input signals are opposite in polarity and the output voltage is proportional to the differential input voltage.
- The common-mode mode of operation of an op-amp is when the input signals are the same in polarity and the output voltage is proportional to the common-mode input voltage.
- The common-mode rejection ratio (CMRR) of an op-amp is a measure of how well the op-amp amplifies the differential input voltage and rejects the common-mode input voltage.
- The CMRR is defined as the ratio of the differential voltage gain to the common-mode voltage gain, denoted by  $CMRR = A_d/A_{cm}$ .
- The ideal op-amp has an infinite CMRR, meaning that it amplifies only the differential input voltage and ignores the common-mode input voltage.
- The practical op-amp has a finite CMRR, meaning that it amplifies both the differential and the common-mode input voltages, but the common-mode voltage gain is much smaller than the differential voltage gain.
- The common-mode and differential mode signals are associated with both



op-amps and interference noise in circuits.

- Common-mode interference, or noise, is the unwanted voltage that is the same for signals applied to both inputs of an op-amp.
- Differential-mode interference, or noise, is the unwanted voltage that is opposite for signals applied to both inputs of an op-amp.
- Common-mode interference can be caused by external sources, such as power supply fluctuations, electromagnetic interference, or ground loops.
- Differential-mode interference can be caused by internal sources, such as mismatched resistors, capacitors, or inductors in the circuit.
- Common-mode interference can be reduced by using shielding, filtering, or grounding techniques.
- Differential-mode interference can be reduced by using balanced or twisted-pair wires, or by using differential amplifiers or op-amps.

#### **Comparators for the notes of the Unit 4 - Operational Amplifiers in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING**

- A comparator is a circuit that uses an operational amplifier to compare two voltages and output a high or low signal depending on which voltage is larger .
- A comparator can be used for various applications, such as polarity identification, analog-to-digital conversion, switch driving, waveform generation, and pulse-edge detection .
- A comparator can be configured by using an operational amplifier in its open-loop state, that is, without any feedback resistor. This allows the operational amplifier to have a very high gain and a very fast response time.
- A comparator can have two inputs: a non-inverting input (+) and an inverting input (-). The output voltage is determined by the following rules:
  - If  $V_+ > V_-$ , then the output voltage is equal to the positive supply voltage ( $V_{cc}$ ).
  - If  $V_+ < V_-$ , then the output voltage is equal to the negative supply voltage ( $V_{ee}$ ).
  - If  $V_+ = V_-$ , then the output voltage is undefined and may oscillate between  $V_{cc}$  and  $V_{ee}$ .
- A comparator can have different types of outputs, such as open-collector, push-pull, or rail-to-rail. The output type affects the load driving capability and the power consumption of the comparator.
- A comparator can also have hysteresis, which is a small difference between the switching thresholds of the comparator. Hysteresis can prevent false triggering and noise interference in the comparator output .
- A comparator can be used to form a window comparator, which is a circuit that checks whether the input voltage is within a certain range or window of values. A window comparator uses two comparators and two reference

voltages to produce a two-state output.

## Unit 5 - Digital Electronics

- Digital electronics is the branch of electronics that deals with the representation and manipulation of data in digital form.
- Digital electronics involves the study of digital signals and the engineering of devices that use or produce them.
- Digital electronic circuits are usually made from large assemblies of logic gates, often packaged in integrated circuits.
- Logic gates are the basic building blocks of all digital electronic circuits. They perform logical operations on one or more binary inputs and produce a single binary output.
- There are three types of logic gates: AND, OR, and NOT. Each gate has a symbol, a truth table, and a Boolean expression that describes its function.
- AND gate: The output is 1 only when both inputs are 1. Symbol: AND gate Truth table:

| A | B | Output |
|---|---|--------|
| 0 | 0 | 0      |
| 0 | 1 | 0      |
| 1 | 0 | 0      |
| 1 | 1 | 1      |

Boolean expression: Output = A.B

- OR gate: The output is 1 when either or both inputs are 1. Symbol: OR gate Truth table:

| A | B | Output |
|---|---|--------|
| 0 | 0 | 0      |
| 0 | 1 | 1      |
| 1 | 0 | 1      |
| 1 | 1 | 1      |

Boolean expression: Output = A+B

- NOT gate: The output is the complement of the input. Symbol: NOT gate Truth table:

| A | Output |
|---|--------|
| 0 | 1      |

| A | Output |
|---|--------|
| 1 | 0      |

Boolean expression: Output = A'

- Other types of logic gates, such as NAND, NOR, XOR, and XNOR, can be derived from the combination of the basic gates.
- Boolean algebra is the mathematical system that deals with the manipulation of binary variables and logic expressions.
- Boolean algebra has some properties, such as commutativity, associativity, distributivity, identity, complement, and duality, that can be used to simplify logic expressions.
- Minimization of Boolean functions is the process of finding the simplest or most efficient way to represent a logic function.
- There are different methods of minimization, such as algebraic method, Karnaugh map, and Quine-McCluskey method.
- Karnaugh map is a graphical technique that can be used to simplify logic functions of up to four variables by grouping adjacent cells that have the same output.
- Canonical and standard forms are the ways of representing logic functions in a unique and unambiguous manner.
- There are two types of canonical forms: sum of products (SOP) and product of sums (POS).
- There are two types of standard forms: minterm and maxterm.
- Minterm is a product term that contains all the variables of the function, either in complemented or uncomplemented form.
- Maxterm is a sum term that contains all the variables of the function, either in complemented or uncomplemented form.
- Functional completeness is the property of a set of logic gates that can be used to implement any logic function.
- The set of {AND, OR, NOT} gates is functionally complete, as any logic function can be expressed using these gates.
- The set of {NAND} or {NOR} gates is also functionally complete, as any logic function can be expressed using only one type of these gates.
- Digital electronics has many applications, such as computers, microcontrollers, calculators, digital clocks, communication systems, encryption, and artificial intelligence .
- Digital electronics

#### **Number System and Representation for the Notes of the Unit 5 - Digital Electronics in the Subject of Fundamentals of Electronics Engineering**

- A number system is a way of representing information using symbols or digits.

- The base or radix of a number system is the total number of symbols or digits used in the number system.
- The most common number systems in digital electronics are decimal, binary, octal, and hexadecimal.
- Decimal number system uses 10 symbols or digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. It is the standard system for human arithmetic and communication.
- Binary number system uses 2 symbols or digits: 0 and 1. It is the simplest and most fundamental system for digital electronics and computers. Each digit or bit can represent two states: on or off, true or false, high or low, etc.
- Octal number system uses 8 symbols or digits: 0, 1, 2, 3, 4, 5, 6, and 7. It is a convenient way of grouping binary bits into three-bit groups or octets. Each octal digit represents three binary digits.
- Hexadecimal number system uses 16 symbols or digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, and F. It is another convenient way of grouping binary bits into four-bit groups or nibbles. Each hexadecimal digit represents four binary digits.
- To convert a number from one base to another, there are two methods: division-remainder method and multiplication method.
- Division-remainder method is used to convert a number from a higher base to a lower base. The steps are:
  - Divide the number by the base of the target system.
  - Write down the remainder as the rightmost digit of the result.
  - Repeat the process with the quotient until it becomes zero.
  - Write down the final quotient as the leftmost digit of the result.
  - The result is the number in the target system.
- Multiplication method is used to convert a number from a lower base to a higher base. The steps are:
  - Multiply the number by the base of the target system.
  - Write down the integer part of the product as the leftmost digit of the result.
  - Repeat the process with the fractional part of the product until it becomes zero or reaches the desired precision.
  - Write down the final fractional part as the rightmost digit of the result.
  - The result is the number in the target system.
- To convert a number from binary to octal or hexadecimal, or vice versa, there is a shortcut method: grouping method. The steps are:
  - Group the binary digits from right to left into three-bit groups for octal or four-bit groups for hexadecimal.
  - Write down the corresponding octal or hexadecimal digit for each group.
  - The result is the number in the target system.
- To convert a number from octal to hexadecimal, or vice versa, there is another shortcut method: intermediate binary method. The steps are:
  - Convert the number from octal to binary using the grouping method.

- Convert the binary number to hexadecimal using the grouping method.
- The result is the number in the target system.
- In digital electronics, there are two ways of representing numbers: fixed point and floating point.
- Fixed point representation uses a fixed number of bits to represent the integer and fractional parts of a number. It is simple and fast, but has limited range and precision.
- Floating point representation uses a variable number of bits to represent the sign, exponent, and mantissa of a number. It is complex and slow, but has greater range and precision.

### **Binary arithmetic for the notes of the Unit 5 - Digital Electronics in the subject of FUNDAMENTALS OF ELECTRONICS ENGINEERING**

- Binary arithmetic is the process of performing mathematical operations on binary numbers, which are represented by only two digits: 0 and 1.
- Binary arithmetic is essential for all the digital computers and many other digital systems, such as microprocessors, flip flops, counters, etc.
- Binary arithmetic involves four basic operations: addition, subtraction, multiplication, and division.
- Binary addition is the simplest and most fundamental operation of binary arithmetic. It follows four rules:
  - $0 + 0 = 0$
  - $0 + 1 = 1$
  - $1 + 0 = 1$
  - $1 + 1 = 10$  (carry 1 to the next higher bit)
- Binary subtraction is the inverse of binary addition. It follows four rules:
  - $0 - 0 = 0$
  - $0 - 1 = 1$  (borrow 1 from the next higher bit)
  - $1 - 0 = 1$
  - $1 - 1 = 0$
- Binary multiplication is similar to decimal multiplication, except that each digit is either 0 or 1. It follows two rules:
  - $0 \times 0 = 0$
  - $0 \times 1 = 0$
  - $1 \times 0 = 0$
  - $1 \times 1 = 1$
- Binary division is similar to decimal division, except that each digit is either 0 or 1. It follows two rules:

- $0 / 0 = \text{undefined}$
- $0 / 1 = 0$
- $1 / 0 = \text{undefined}$
- $1 / 1 = 1$

- Binary arithmetic can be performed using various methods and techniques, such as binary adders, binary subtractors, binary multipliers, binary dividers, etc. These are electronic circuits that can implement binary arithmetic using voltage signals (high for 1, low for 0).

### Introduction of Basic and Universal Gates

- Logic gates are the fundamental building blocks of digital electronics. They perform basic logical operations on binary inputs and produce a binary output.
- There are three basic logic gates: AND, OR, and NOT. They have the following truth tables and symbols:

| A | B | A AND B |
|---|---|---------|
| 0 | 0 | 0       |
| 0 | 1 | 0       |
| 1 | 0 | 0       |
| 1 | 1 | 1       |

AND gate symbol

| A | B | A OR B |
|---|---|--------|
| 0 | 0 | 0      |
| 0 | 1 | 1      |
| 1 | 0 | 1      |
| 1 | 1 | 1      |

OR gate symbol

| A | NOT A |
|---|-------|
| 0 | 1     |
| 1 | 0     |

NOT gate symbol

- Universal gates are logic gates that can be used to implement any other logic gate or Boolean function. They are NAND and NOR gates. They

have the following truth tables and symbols:

| A | B | A NAND B |
|---|---|----------|
| 0 | 0 | 1        |
| 0 | 1 | 1        |
| 1 | 0 | 1        |
| 1 | 1 | 0        |

NAND gate symbol

| A | B | A NOR B |
|---|---|---------|
| 0 | 0 | 1       |
| 0 | 1 | 0       |
| 1 | 0 | 0       |
| 1 | 1 | 0       |

NOR gate symbol

- The reason why NAND and NOR gates are universal is that they can be combined to produce any other logic gate or Boolean function. For example, the following diagrams show how to construct an AND gate, an OR gate, and a NOT gate using only NAND gates:

AND gate using NAND gates

OR gate using NAND gates

NOT gate using NAND gate

- Similarly, the following diagrams show how to construct an AND gate, an OR gate, and a NOT gate using only NOR gates:

AND gate using NOR gates

OR gate using NOR gates

NOT gate using NOR gate

- Digital logic circuits are integrated into a single IC to design several processors and controllers. The IC number of a logic gate indicates the type and number of gates in the IC. For example, the IC number 7400 indicates a quad 2-input NAND gate, which

### Using Boolean algebra simplification of Boolean function

- Boolean algebra is a mathematical system that deals with binary variables and logic operations.

- Boolean functions are expressions that use binary variables and logic operations to produce a binary output.
- Simplification of Boolean functions means reducing the number of terms and/or operations in a Boolean function, which leads to simpler and cheaper implementations of logic circuits.
- Simplification of Boolean functions can be done by using the theorems and rules of Boolean algebra, such as:
  - Identity laws:  $X + 0 = X$ ,  $X \cdot 1 = X$
  - Complement laws:  $X + X' = 1$ ,  $X \cdot X' = 0$
  - Commutative laws:  $X + Y = Y + X$ ,  $X \cdot Y = Y \cdot X$
  - Associative laws:  $(X + Y) + Z = X + (Y + Z)$ ,  $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$
  - Distributive laws:  $X \cdot (Y + Z) = (X \cdot Y) + (X \cdot Z)$ ,  $X + (Y \cdot Z) = (X + Y) \cdot (X + Z)$
  - De Morgan's laws:  $(X + Y)' = X' \cdot Y'$ ,  $(X \cdot Y)' = X' + Y'$
  - Absorption laws:  $X + (X \cdot Y) = X$ ,  $X \cdot (X + Y) = X$
  - Redundancy laws:  $X + X = X$ ,  $X \cdot X = X$
  - Consensus laws:  $X \cdot Y + X' \cdot Z + Y \cdot Z = X \cdot Y + X' \cdot Z$ ,  $X + Y \cdot Z + X' \cdot Y = X + Y \cdot Z$
- Simplification of Boolean functions can be done by applying these rules in a step-by-step manner, until no further simplification is possible.
- Example: Simplify the Boolean function  $F = A \cdot B + A \cdot B' + B \cdot C$ 
  - Step 1: Apply the distributive law to factor out A:  $F = A \cdot (B + B') + B \cdot C$
  - Step 2: Apply the complement law to simplify  $B + B'$ :  $F = A \cdot 1 + B \cdot C$
  - Step 3: Apply the identity law to eliminate 1:  $F = A + B \cdot C$
  - Step 4: No further simplification is possible, so the final answer is  $F = A + B \cdot C$

## K Map Minimization upto 6 Variables

- Karnaugh map (K-map) is a graphical tool to simplify a Boolean expression or to convert a truth table to its corresponding logic circuit in a simple and orderly manner.
- K-map can handle up to four variables for SOP (sum of product) and POS (product of sum) expressions. For more than four variables, Quine-McCluskey method is preferred.
- However, it is possible to extend the K-map method to handle up to six variables by using three-dimensional or multidimensional K-maps.
- A three-dimensional K-map consists of two or more two-dimensional K-maps stacked on top of each other, with the corresponding cells connected by dotted lines.



- A multidimensional K-map consists of a rectangular array of two-dimensional K-maps, with the corresponding cells connected by solid or dashed lines.
- The rules for grouping and minimizing the K-map cells are the same as for the two-dimensional case, except that the groups can span across the layers or dimensions of the K-map.
- The following are some examples of K-map minimization for up to six variables:

**Example 1: Minimize the following SOP expression using a three-dimensional K-map:**

$$F(A,B,C,D,E) = m(0,1,2,3,4,5,8,9,10,11,12,13,16,17,18,19,20,21,24,25,26,27,28,29)$$

- The expression has five variables, so we need a three-dimensional K-map with two layers of 4x4 two-dimensional K-maps.
- The first layer corresponds to  $E=0$  and the second layer corresponds to  $E=1$ .
- The rows and columns of each layer are labeled by the values of A and B, and C and D, respectively.
- The cells are filled with 1s for the minterms of the expression and 0s for the remaining cells.
- The K-map is shown below:

| E=0 | CD | 00 | 01 | 11 | 10 |
|-----|----|----|----|----|----|
| AB  | 00 | 1  | 1  | 1  | 1  |
|     | 01 | 1  | 1  | 1  |    |
|     | 11 | 0  | 0  | 0  | 0  |
|     | 10 | 1  | 1  | 1  | 1  |

| E=1 | CD | 00 | 01 | 11 | 10 |
|-----|----|----|----|----|----|
| AB  | 00 | 1  | 1  | 1  | 1  |
|     | 01 | 1  | 1  | 1  |    |
|     | 11 | 0  | 0  | 0  | 0  |
|     | 10 | 1  | 1  | 1  | 1  |

- The groups of adjacent 1s are shown by different colors and shapes, and the corresponding prime implicants are:

| Group         | Prime Implicant |
|---------------|-----------------|
| Red Square    | $A'B'E'$        |
| Green Circle  | $A'B'E$         |
| Blue Triangle | $AB'E'$         |

| Group          | Prime Implicant |
|----------------|-----------------|
| Yellow Diamond | AB'E            |
| Pink Hexagon   | C'D'E'          |
| Cyan Star      | C'D'E           |

- The minimum SOP expression is obtained by selecting the essential prime implicants and the minimum number of remaining prime implicants to cover all the 1s in the K-map.
- The essential prime implicants are those that cover at least one 1 that is not covered by any other prime implicant.
- In this case, the essential prime implicants are the red square, the green circle, the blue triangle, and the yellow diamond, as they cover the corners of the K-map that are not covered by any other group.
- The remaining prime implicants are the pink hexagon and the cyan star, which cover the middle cells of the K-map that are also covered by other groups.
- We can select either one of them to complete the cover, as they are equivalent in terms of cost and literals.
- Let us choose the pink hexagon for simplicity.
- The minimum SOP expression is:

$$F(A,B,C,D,E) = A'B'E' + A'B'E + AB'E' + AB'E + C'D'E'$$

## Unit 6 - Fundamentals of Communication Engineering

- Communication engineering is a field that focuses on supporting systems that transfer information from one place to another.
- Information can be in the form of signals, such as audio, video, data, or text.
- Communication systems can be classified into analog or digital, depending on the nature of the signals and the devices used to process them.
- Some of the basic communication engineering concepts are:
  - Signals, spectra, and bandwidth: Signals are the physical representation of information, spectra are the frequency components of signals, and bandwidth is the range of frequencies occupied by a signal.
  - Attenuation, distortion, and noise: Attenuation is the loss of signal power due to transmission medium, distortion is the alteration of signal shape or quality due to non-linear effects, and noise is the unwanted interference that affects the signal quality.
  - Filtering, equalizing, and companding: Filtering is the process of removing unwanted frequency components from a signal, equalizing is the process of compensating for the distortion caused by the transmission medium, and companding is the process of compressing and expanding the dynamic range of a signal to reduce the effect of noise.
  - Modulation and multiplexing: Modulation is the process of changing

the characteristics of a carrier signal according to the information signal, and multiplexing is the process of combining multiple signals into one signal for transmission over a common medium.

- Information measurement, source and channel coding, channel capacity and Shannon theorems: Information measurement is the quantification of the amount of information in a signal, source coding is the process of reducing the redundancy in a signal, channel coding is the process of adding redundancy to a signal to improve its reliability, channel capacity is the maximum rate of information that can be transmitted over a channel, and Shannon theorems are the fundamental limits of communication systems.
- Baseband data transmission, digital modulation and spread-spectrum systems: Baseband data transmission is the transmission of digital signals without modulation, digital modulation is the modulation of digital signals using discrete values of carrier signal, and spread-spectrum systems are the systems that use a wide bandwidth to transmit signals with low power and high security.
- Local area networks and architecture and performance: Local area networks are the networks that connect devices within a limited geographical area, such as a building or a campus, and their architecture and performance depend on the topology, medium, protocol, and traffic characteristics of the network.
- Layered network architecture, protocols, switching techniques, TCP/IP, traffic engineering and basic capacity analysis: Layered network architecture is the division of network functions into different layers, such as physical, data link, network, transport, and application, protocols are the rules and procedures that govern the communication between network devices, switching techniques are the methods of transferring data from one device to another, such as circuit switching, packet switching, and message switching, TCP/IP is the most widely used protocol suite for internet communication, traffic engineering is the optimization of network performance and resource utilization, and basic capacity analysis is the estimation of the maximum amount of data that can be transmitted over a network.

### **Basics of signal representation and analysis**

- A signal is a physical quantity that varies with time, space, or any other independent variable. It can carry information about the state or behavior of a system or phenomenon. Examples of signals are voltage, current, sound, light, temperature, etc.
- Signal representation is the process of describing a signal in terms of mathematical functions or symbols. Signal representation can be done in different domains, such as time domain, frequency domain, or other transform domains.

- Signal analysis is the process of extracting useful information from a signal, such as its amplitude, frequency, phase, spectrum, energy, power, etc. Signal analysis can be done using various techniques, such as Fourier analysis, Laplace transform, Z-transform, wavelet transform, etc.
- Time domain representation is the most natural and intuitive way of representing a signal. It shows how the signal varies with time. Time domain representation is useful for studying the transient or dynamic behavior of a signal.
- Frequency domain representation is another way of representing a signal. It shows how the signal is composed of different frequency components. Frequency domain representation is useful for studying the periodic or steady-state behavior of a signal. Frequency domain representation can be obtained by applying the Fourier transform to the time domain representation.
- Other transform domains are alternative ways of representing a signal. They can provide different perspectives or insights into the signal characteristics. Some examples of other transform domains are Laplace domain, Z-domain, wavelet domain, etc. Each transform domain has its own advantages and limitations.

### **Electromagnetic spectrum**

- The electromagnetic spectrum is the range of all types of electromagnetic radiation, which are energy waves that travel and spread out as they go .
- Electromagnetic radiation can be classified by its frequency, wavelength, or photon energy, which are inversely proportional to each other .
- The electromagnetic spectrum covers electromagnetic waves with frequencies ranging from below one hertz to above  $10^{25}$  hertz, corresponding to wavelengths from thousands of kilometers down to a fraction of the size of an atomic nucleus.
- The electromagnetic spectrum is divided into different regions based on the properties and applications of the electromagnetic waves. The regions are, from lowest to highest frequency: radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, and gamma rays .
- Radio waves have the lowest frequency and longest wavelength, and are used for communication, broadcasting, and navigation .
- Microwaves have higher frequency and shorter wavelength than radio waves, and are used for heating, radar, and satellite communication .
- Infrared waves have higher frequency and shorter wavelength than microwaves, and are emitted by warm objects. They are used for thermal imaging, remote sensing, and night vision .
- Visible light waves have higher frequency and shorter wavelength than

infrared waves, and are the only electromagnetic waves that humans can see. They are used for vision, photography, and illumination .

- Ultraviolet waves have higher frequency and shorter wavelength than visible light waves, and are emitted by the sun and other hot objects. They are used for sterilization, fluorescence, and sun tanning, but can also cause skin damage and cancer .
- X-rays have higher frequency and shorter wavelength than ultraviolet waves, and are produced by high-energy processes such as atomic collisions and electron acceleration. They are used for medical imaging, security scanning, and crystallography, but can also cause tissue damage and mutations .
- Gamma rays have the highest frequency and shortest wavelength of all electromagnetic waves, and are generated by nuclear reactions and cosmic events. They are used for nuclear medicine, astronomy, and radiation therapy, but can also be lethal to living organisms .
- The electromagnetic spectrum is shown in the following diagram, where the frequency increases from left to right and the wavelength decreases from left to right :

electromagnetic spectrum diagram

### Elements of a Communication System

A communication system is a system that enables the exchange of information between two or more points. It consists of the following basic elements :

- **Information source:** This is the origin of the message or the data that needs to be communicated. It can be a person, a device, or a process that generates the information. Examples are a microphone, a camera, a keyboard, etc.
- **Input transducer:** This is a device that converts the information from the source into a form suitable for transmission. It can be an analog or a digital device that transforms the information into electrical signals. Examples are a microphone, a camera, a keyboard, etc.
- **Transmitter:** This is a device that modulates the electrical signals from the input transducer and amplifies them for transmission over a channel. It can use different techniques of modulation, such as amplitude modulation, frequency modulation, phase modulation, etc. Examples are a radio transmitter, a telephone transmitter, a satellite transmitter, etc.
- **Channel:** This is the medium that carries the modulated signals from the transmitter to the receiver. It can be a wired or a wireless medium that has certain characteristics, such as bandwidth, attenuation, noise, interference, etc. Examples are a coaxial cable, a fiber optic cable, a radio wave, a microwave, etc.
- **Receiver:** This is a device that demodulates the signals from the channel and amplifies them for output. It can use different techniques of demodulation, such as envelope detection, frequency discrimination, phase detec-

tion, etc. Examples are a radio receiver, a telephone receiver, a satellite receiver, etc.

- **Output transducer:** This is a device that converts the electrical signals from the receiver into a form suitable for the destination. It can be an analog or a digital device that transforms the signals into information. Examples are a speaker, a monitor, a printer, etc.
- **Destination:** This is the final point of the communication system where the information is delivered. It can be a person, a device, or a process that consumes the information. Examples are a speaker, a monitor, a printer, etc.

The communication system can be classified into different types based on the nature of the information, the mode of transmission, the direction of communication, etc. Some of the common types are:

- **Analog communication system:** This is a system that uses analog signals to transmit and receive information. Analog signals are continuous signals that vary in amplitude and frequency. Examples are radio, television, telephone, etc.
- **Digital communication system:** This is a system that uses digital signals to transmit and receive information. Digital signals are discrete signals that have only two values, 0 and 1. Examples are computer networks, cellular phones, digital television, etc.
- **Simplex communication system:** This is a system that allows communication in one direction only. The transmitter can only send information and the receiver can only receive information. Examples are radio broadcasting, television broadcasting, etc.
- **Duplex communication system:** This is a system that allows communication in both directions, but not at the same time. The transmitter and the receiver can switch roles and take turns to send and receive information. Examples are walkie-talkies, telephones, etc.
- **Half-duplex communication system:** This is a system that allows communication in both directions, but only one at a time. The transmitter and the receiver can send and receive information simultaneously, but they have to share the same channel. Examples are push-to-talk radios, intercoms, etc.
- **Full-duplex communication system:** This is a system that allows communication in both directions, and at the same time. The transmitter and the receiver can send and receive information simultaneously, and they have separate channels. Examples are cellular phones, internet, etc.

## Need of Modulation and Typical Applications

Modulation is the process of changing the characteristics of a carrier wave (such as amplitude, frequency, or phase) according to the information contained in a message signal. Modulation is essential for various reasons, such as:

- To increase the range and quality of communication: Modulation allows the transmission of low-frequency message signals over long distances by using high-frequency carrier waves. This reduces the attenuation and noise interference of the signal. Modulation also improves the quality of the signal by reducing the distortion and fading effects.
- To facilitate multiplexing: Modulation enables the transmission of multiple message signals over the same channel by using different carrier frequencies or modulation schemes. This increases the efficiency and capacity of the communication system.
- To enable wireless communication: Modulation eliminates the need for wires or cables in the communication system by using electromagnetic waves as the carrier. Modulation allows the use of various wireless devices, such as radio, television, mobile phones, satellites, etc.
- To encrypt and secure the information: Modulation can be used to encode and decode the message signal using a secret key or a code. This prevents unauthorized access or interception of the information.

Some of the typical applications of modulation are:

- Amplitude modulation (AM): This is the simplest form of modulation, where the amplitude of the carrier wave is varied according to the message signal. AM is used for broadcasting radio signals, transmitting voice and music, and sending data over telephone lines.
- Frequency modulation (FM): This is the modulation where the frequency of the carrier wave is varied according to the message signal. FM is used for high-fidelity audio transmission, such as FM radio, television sound, and mobile phones.
- Phase modulation (PM): This is the modulation where the phase of the carrier wave is varied according to the message signal. PM is used for digital communication, such as binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), and quadrature amplitude modulation (QAM).
- Pulse modulation (PM): This is the modulation where the carrier wave is a series of pulses, and the message signal is represented by the presence or absence of the pulses. PM is used for transmitting digital signals, such as pulse amplitude modulation (PAM), pulse width modulation (PWM), pulse position modulation (PPM), and pulse code modulation (PCM).

### **Fundamentals of amplitude modulation and demodulation techniques**

- Amplitude modulation (AM) is a technique to transmit information via radio carrier waveform by varying the amplitude of the carrier signal in proportion to the amplitude of the modulation signal that is to be transmitted.
- The modulation signal can be an audio signal, a video signal, or any other type of signal that carries information.
- The carrier signal is usually a high-frequency sinusoidal wave that can be

easily radiated by an antenna and propagated through the air or space.

- The modulated signal has the same frequency as the carrier signal, but its amplitude varies according to the modulation signal.
- The modulated signal can be represented by the following equation:

$$s(t) = A_c[1 + k_a m(t)] \cos(2\pi f_c t)$$

where  $s(t)$  is the modulated signal,  $A_c$  is the amplitude of the carrier signal,  $k_a$  is the amplitude modulation index,  $m(t)$  is the modulation signal, and  $f_c$  is the frequency of the carrier signal.

- The amplitude modulation index  $k_a$  is a measure of the degree of modulation and is defined as the ratio of the maximum amplitude of the modulation signal to the amplitude of the carrier signal:

$$k_a = \frac{A_m}{A_c}$$

where  $A_m$  is the maximum amplitude of the modulation signal.

- The amplitude modulation index can vary from 0 to 1. When  $k_a = 0$ , there is no modulation and the modulated signal is equal to the carrier signal. When  $k_a = 1$ , the modulation is 100% and the modulated signal has the maximum possible amplitude variation.
- The modulated signal can be analyzed in the frequency domain using the Fourier transform. The modulated signal has three frequency components: the carrier frequency  $f_c$ , the lower sideband frequency  $f_c - f_m$ , and the upper sideband frequency  $f_c + f_m$ , where  $f_m$  is the frequency of the modulation signal.
- The modulated signal can be represented by the following equation in the frequency domain:

$$S(f) = \frac{A_c}{2} \delta(f - f_c) + \frac{A_c k_a}{4} [M(f - f_c) + M(f + f_c)]$$

where  $S(f)$  is the modulated signal in the frequency domain,  $\delta(f)$  is the Dirac delta function, and  $M(f)$  is the modulation signal in the frequency domain.

- The modulated signal occupies a bandwidth of  $2f_m$  around the carrier frequency. The bandwidth of the modulated signal is twice the bandwidth of the modulation signal.
- Amplitude demodulation is the process of recovering the modulation signal from the modulated signal. There are different techniques for amplitude demodulation, such as envelope detection, synchronous detection, and product detection.



- Envelope detection is the simplest and most common technique for amplitude demodulation. It uses a diode and a capacitor to extract the envelope of the modulated signal, which is proportional to the modulation signal.
- The envelope detector circuit consists of a diode, a capacitor, and a resistor. The diode rectifies the modulated signal and allows only the positive half cycles to pass. The capacitor charges to the peak value of the rectified signal and discharges through the resistor. The output of the envelope detector is the voltage across the capacitor, which follows the envelope of the modulated signal.
- The envelope detector works well when the modulation index is high and the carrier frequency is much higher than the modulation frequency. However, when the modulation index is low or the carrier frequency is close to the modulation frequency, the envelope detector may fail to track the envelope of the modulated signal and produce distortion.
- Synchronous detection is a more accurate and complex technique for amplitude demodulation. It uses a local oscillator that generates a carrier signal with the same frequency and phase as the original carrier signal. The modulated signal and the local oscillator signal are multiplied by a mixer, which produces the sum and difference of the two signals. The output of the mixer is then filtered by a low-pass filter, which removes the sum signal and passes the difference signal, which is equal to the modulation signal.
- The synchronous detector works well for any modulation index and carrier frequency, but it requires a precise and stable local oscillator that can match the frequency and phase of the original carrier signal. This may be difficult to achieve in practice, especially for high-frequency signals.
- Product detection

## **Introduction to Wireless Communication**

- Wireless communication is the transmission of voice and data without cable or wires.
- Wireless communication uses electromagnetic signals, such as radio waves, to broadcast information from sending facilities to intermediate and end-user devices.
- Wireless communication has many advantages over wired communication, such as mobility, scalability, flexibility, and cost-effectiveness.
- Wireless communication also has some challenges, such as interference, security, reliability, and power consumption.
- Wireless communication can be classified into different types based on the range, frequency, topology, and application of the wireless network, such as cellular, satellite, Wi-Fi, Bluetooth, RFID, and NFC.
- Wireless communication has many applications in various fields, such as

telecommunication, entertainment, education, health care, transportation, and military.

- Wireless communication is a rapidly evolving field that requires constant innovation and research to meet the growing demands of users and society.

### Overview of wireless communication

- Wireless communication is the transfer of information (telecommunication) between two or more points without the use of an electrical conductor, optical fiber or other continuous guided medium for the transfer .
- Wireless communication generally works through electromagnetic signals that are broadcast by an enabled device within the air, physical environment or atmosphere. The sending device can be a sender or an intermediate device with the ability to propagate wireless signals .
- Wireless communication can be classified into different types based on the range, frequency, modulation, multiplexing, and application of the signals. Some of the common types are:
  - Radio communication: It uses radio waves to transmit information over long distances. Radio communication can be further divided into broadcast radio, amateur radio, cellular radio, satellite radio, etc.
  - Microwave communication: It uses microwaves to transmit information over short distances. Microwave communication can be further divided into terrestrial microwave, satellite microwave, radar, etc.
  - Infrared communication: It uses infrared light to transmit information over short distances. Infrared communication can be further divided into remote control, wireless LAN, optical wireless, etc.
  - Optical communication: It uses visible light or ultraviolet light to transmit information over short distances. Optical communication can be further divided into free-space optical, laser communication, visible light communication, etc.
- Wireless communication can also be classified into different types based on the topology, architecture, and protocol of the network. Some of the common types are:
  - Wireless personal area network (WPAN): It is a network that connects devices within a short range, typically within a person's reach. Examples of WPAN are Bluetooth, Zigbee, NFC, etc.
  - Wireless local area network (WLAN): It is a network that connects devices within a local area, typically within a building or a campus. Examples of WLAN are Wi-Fi, WiMAX, etc.
  - Wireless metropolitan area network (WMAN): It is a network that connects devices within a metropolitan area, typically within a city or a region. Examples of WMAN are 4G, 5G, etc.
  - Wireless wide area network (WWAN): It is a network that connects devices across a wide area, typically across a country or a continent.

Examples of WWAN are cellular networks, satellite networks, etc.

- Wireless communication has many advantages over wired communication, such as mobility, scalability, flexibility, cost-effectiveness, etc. However, wireless communication also faces many challenges, such as interference, security, reliability, power consumption, etc. Therefore, wireless communication requires careful design, implementation, and management to ensure its performance and quality.

## Cellular Communication

Cellular communication is a form of wireless communication that uses radio waves to transmit and receive signals from mobile devices. Cellular communication is based on the concept of dividing a geographical area into small cells, each with its own base station and frequency channel. Cellular communication allows multiple users to share the same frequency spectrum without interfering with each other.

Some of the topics covered in cellular communication are:

- **Cellular system architecture:** This topic explains the components and functions of a cellular system, such as mobile stations, base stations, mobile switching centers, and control channels. It also describes the types of cellular systems, such as analog, digital, and hybrid systems.
- **Cellular system design:** This topic covers the principles and methods of cellular system design, such as frequency reuse, cell splitting, sectoring, and handoff. It also discusses the factors that affect the system capacity and performance, such as interference, noise, fading, and shadowing.
- **Cellular system operation:** This topic covers the procedures and protocols of cellular system operation, such as call initiation, call termination, call processing, and roaming. It also explains the features and services of cellular systems, such as voice, data, SMS, and multimedia.
- **Cellular system standards:** This topic covers the various standards and technologies of cellular systems, such as AMPS, GSM, CDMA, TDMA, and LTE. It also compares the advantages and disadvantages of different standards and technologies in terms of spectrum efficiency, data rate, security, and compatibility.

## Different Generations and Standards in Cellular Communication Systems

- Cellular communication systems are wireless networks that use radio waves to transmit voice and data signals over a large area.
- Cellular communication systems are classified into different generations based on their technology, features, and standards.
- The main generations of cellular communication systems are:

- **1G (First Generation):** The first generation of cellular systems used analog modulation and frequency division multiple access (FDMA) to provide voice-only services. The main standards of 1G were Advanced Mobile Phone System (AMPS) in North America, Total Access Communication System (TACS) in Europe, and Nordic Mobile Telephone (NMT) in Scandinavia. 1G systems had low capacity, poor voice quality, and no security or encryption .
- **2G (Second Generation):** The second generation of cellular systems used digital modulation and multiple access techniques such as time division multiple access (TDMA), code division multiple access (CDMA), and frequency hopping spread spectrum (FHSS) to provide voice and data services. The main standards of 2G were Global System for Mobile Communications (GSM), Interim Standard 95 (IS-95), and Personal Digital Cellular (PDC). 2G systems had higher capacity, better voice quality, and improved security and encryption .
- **3G (Third Generation):** The third generation of cellular systems used wideband CDMA (W-CDMA) and orthogonal frequency division multiplexing (OFDM) to provide high-speed data and multimedia services. The main standards of 3G were Universal Mobile Telecommunications System (UMTS), CDMA2000, and High-Speed Packet Access (HSPA). 3G systems had higher data rates, better spectral efficiency, and enhanced quality of service (QoS) and mobility .
- **4G (Fourth Generation):** The fourth generation of cellular systems used OFDM and multiple-input multiple-output (MIMO) to provide very high-speed data and multimedia services. The main standards of 4G were Long Term Evolution (LTE), WiMAX, and LTE-Advanced. 4G systems had higher data rates, lower latency, and improved scalability and interoperability .
- **5G (Fifth Generation):** The fifth generation of cellular systems is still under development and aims to provide ultra-high-speed data and multimedia services with low latency, high reliability, and massive connectivity. The main technologies of 5G are millimeter wave (mmWave), massive MIMO, beamforming, and network slicing. 5G systems are expected to support new applications such as virtual reality, augmented reality, and Internet of Things (IoT) .

## Fundamentals of Satellite & Radar Communication

Satellite and radar communication are two types of wireless communication systems that use electromagnetic waves to transmit and receive information. They have different applications, advantages and disadvantages, and design principles.

### Satellite Communication

- Satellite communication involves four steps:
  - An uplink Earth station or other ground equipment transmits the desired signal to the satellite
  - The satellite amplifies the incoming signal and changes the frequency
  - The satellite transmits the signal back to Earth
  - The ground equipment receives the signal
- Satellites are relay stations in space for the transmission of voice, video and data communications.
- Satellites are ideally suited to meet the global communications requirements of military, government and commercial organizations because they provide economical, scalable and highly reliable transmission services that easily reach multiple sites over large geographic areas.
- Satellites can be classified by their orbit, frequency band, function, and generation.
  - Orbit: geostationary (GEO), medium Earth orbit (MEO), low Earth orbit (LEO), etc.
  - Frequency band: C-band, Ku-band, Ka-band, etc.
  - Function: direct broadcast satellites (DBS), high throughput satellites (HTS), very small aperture terminals (VSAT), etc.
  - Generation: first generation (passive reflectors), second generation (active repeaters), third generation (on-board processing), etc.
- Satellite communication systems have some advantages and disadvantages:
  - Advantages: wide coverage, flexibility, mobility, reliability, security, etc.
  - Disadvantages: propagation delay, attenuation, interference, noise, orbital congestion, etc.

## Radar Communication

- Radar communication involves three steps:
  - A transmitter that produces an electromagnetic signal which is radiated into space by an antenna
  - When this signal strikes an object, it gets reflected or reradiated in many directions
  - A receiver that detects the echo or reflected signal from the objects
- Radar stands for Radio Detection and Ranging System.
- Radar is basically an electromagnetic system used to detect the location and distance of an object from the point where the radar is placed.
- Radar can be classified by their frequency, waveform, modulation, polarization, and function.
  - Frequency: microwave, millimeter wave, etc.
  - Waveform: continuous wave (CW), pulsed, frequency modulated (FM), etc.
  - Modulation: amplitude, frequency, phase, etc.
  - Polarization: linear, circular, elliptical, etc.
  - Function: surveillance, tracking, navigation, mapping, etc.

- Radar communication systems have some advantages and disadvantages:
  - Advantages: all-weather, long-range, high-resolution, high-accuracy, etc.
  - Disadvantages: high-power, high-cost, complex, vulnerable to jamming, clutter, etc.